

Efficient Data Structures for Single-Range-Cover Search with Multi-Dimensional Range Queries

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Abstract

Extracting exact and complete information using range queries is not only time inefficient, on average, but also susceptible to privacy breaches. As a remedy, Demertzis et al. [12] (SIGMOD'16) introduced, in the context of searchable symmetric encryption (SSE), the principle of covering a query with a single canonical range using a DAG structure (TDAG), later formalized by Falzon et al. [15] (VLDB'23) as the single range cover (SRC) search algorithm: a faster and potentially privacy-preserving search principle, extended to multi-dimensional range queries. SRC-search can, however, produce false positives, whose expected number has not been formally analyzed for multi-dimensional index structures. In this work we study how a family of specific Directed Acyclic Graphs (DAGs) could substantially increase the accuracy ratio for two-dimensional rectangle range queries, without asymptotic increase of SRC-search time and storage, compared to the classical search structures for multi-dimensional data, 2D-Trees. The studied structures, called c -2D-DAGs, for $c = 2^\alpha + 1$ with $\alpha \geq 1$, are tunable augmentations of 2D-Trees with denser overlapping branches. We perform a competitive analysis of a c -2D-DAG with respect to the 2D-Tree and derive only an **additive constant returned level overhead** while reaching a substantial **multiplicative nearly-linear increase in accuracy ratio** on average. We also provide a generic machine learning framework to extend our results to non-uniform datasets, and demonstrate its effectiveness on the Gowalla benchmark dataset. Finally, we show that randomizing the choice among equally accurate outputs nodes can give the c -2D-DAG positive entropy of the returned node, which the 2D-Tree lacks, hindering leakage under repeated queries at no cost in accuracy.

Keywords

Searchable Symmetric Encryption (SSE), SRC-Search Algorithms and Data Structures, Multi-Dimensional Range Queries, Directed Acyclic Graph (DAG), Competitive Stochastic Analysis.

1 Introduction

Motivation and applications. Range queries are a widely used primitive in data management, arising in spatial and temporal analytics, geographic information systems, and large-scale data processing [6, 25]. Answering a range query by inspecting every element of a large dataset is prohibitive, so systems instead build an index over the data and resolve each query by descending it. When the data is sensitive, however, the search must not reveal the queried records or the queries themselves to the party performing it, which holds whether the dataset is stored locally or delegated to an external service. Searchable symmetric encryption (SSE) meets this requirement: a client outsources an

encrypted database and later retrieves, via encrypted search tokens, the records matching a query, without revealing the query or the data beyond controlled leakage. Song et al. [24] introduced the problem; Curtmola et al. [11] gave the standard formalization, an encrypted inverted index together with adaptive semantic-security definitions still used as the field's reference notion; and Chase and Kamara [9] generalized this index-based approach to arbitrary structured, labeled data under the name structured encryption. A query is processed in two rounds of communication: in the first round the server searches an encrypted index structure and returns the identifiers of the matching documents; in the second round the client retrieves and decrypts the corresponding encrypted data [16, 17], a step orthogonal to the index search [12, 13, 18], although record retrieval itself affects the system-wide security of the resulting scheme [17, 18]. Supporting range queries efficiently is harder than single-keyword search, since testing every value in a range individually degrades to a linear scan, and the structure of the index itself can leak information about the data or the queries [7, 19, 22, 23]; Faber et al. [14] and Demertzis et al. [12] were among the first to reduce a range query to a small number of token lookups by building the encrypted index over a tree-shaped, range-covering data structure, trading off token count, index size, and false positives via covering rules such as best/uniform/single range cover. Demertzis et al. [13] gave the full treatment of this approach (the TDAG), and Falzon et al. [15] generalized it to a broad class of range-supporting data structures and to multi-attribute range queries, on which we build.

Model, problem, and measures. In range-searchable schemes built on a range-supporting data structure, the client maps a query to a set of canonical nodes of the structure, and the server returns the encrypted data associated with these nodes [15]. We study only the index-retrieval stage: how the index is searched and how tightly the returned canonical range contains the query. Among the covering rules, SRC-search covers a query with a single node, so it may return false positives, whose expected number has not been formally analyzed for multi-dimensional index structures. Our comparison model (Fig. 1) places this study in a concrete setting: a client issues a stream of rectangular range queries, each answered in parallel by two servers indexing the same dataset, one with the 2D-Tree and one with our c -2D-DAG, and we compare the two executions by the additive Returned Level Overhead and the multiplicative Accuracy Ratio.

1.1 Our Contributions

We define the c -2D-DAG family of overlapping, data-dependent structures built on top of the 2D-Tree, and a comparison framework for SRC-search on both these structures with two measures, the Returned Level Overhead and the Accuracy Ratio (Sections 2.1 and 2.2). The framework directly compares the execution of SRC-search on both structures for any stochastic input stream of 2D-range queries (i.e., queries of rectangular shapes).

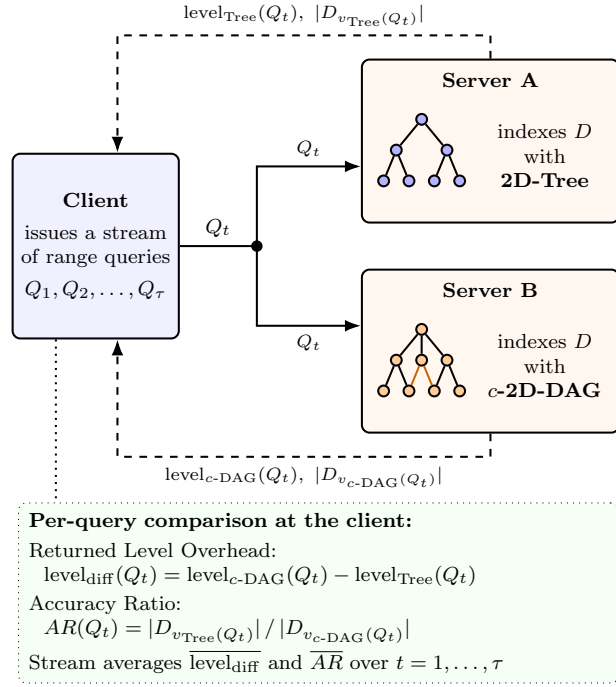


Figure 1: Our comparison model. A client issues a single stream of range queries $Q_1, \dots, Q_\tau$; each query Q_t is forked to two servers indexing the same dataset D , one with the 2D-Tree and one with the c -2D-DAG. Each server runs SRC-search and returns the node it found, i.e., its level and element count; the client compares the two answers to compute the per-query Returned Level Overhead and Accuracy Ratio, and their averages over the stream (Section 2.2).

Under uniform occupancy of a dyadic domain, we formally prove that for every $c = 2^\alpha + 1$, $\alpha \geq 1$, the expected average Returned Level Overhead is below $\frac{(c-2)(5c-3)}{(c-1)^2}$ (equal to 3 for $c = 3$), independently of the dataset size, the query shape, and the stream length, whereas the expected average Accuracy Ratio is $\Omega(\min\{N_0/s_0, N_1/s_1\})$: a constant additive increase in the searching path buys a multiplicative gain in accuracy that grows linearly with the size of the database and with the inverse query size. The results hold for streams mixing several rectangular shapes (Theorem 3.5, Corollary 3.6). Their proofs rest on a detailed analysis of the exact returned level distributions of both structures and of their difference (Lemmas 3.2, 3.3, and 3.4).

We extend these guarantees to arbitrary datasets: the overhead distribution is approximated by a learned uniform baseline, and the bounds carry over with error terms depending on the distance to the baseline and on a level-balance factor of the data (Sec. 4).

Experimental analysis on the Gowalla check-in data ($N = 2^{20}$, four query shapes, 537,000 non-empty queries) confirms two predictions of the theoretical analysis for the 3-2D-DAG: the average Returned Level Overhead stays below the proven bound 3 (ranging from 0.87 to 2.69), and the average Accuracy Ratio (varying from 21.2 to 355.9) exceeds the proven lower bound. It additionally shows that a uniform baseline reproduces the shape of the level overhead distribution on three of the four shapes, while for the remaining shape the empirical distribution is far from the uniform baseline family (Section 5).

We analyze privacy: on the 2D-Tree, exactly one node is returned by SRC-search for a given query, so every execution of

the query discloses the same canonical node to the server. On the c -2D-DAG, several nodes containing a query may be equally accurate; if the search returns one of them uniformly at random, repeated executions of the same query need not disclose the same node. We quantify this uncertainty by the entropy of the returned node: it is zero on the 2D-Tree, at most 2α bits on the c -2D-DAG in the uniform model, and positive on 6.4% to 20.5% of the Gowalla queries, at at no cost in accuracy, since all equally accurate nodes hold the same number of elements (Section 6).

To summarize, the c -2D-DAGs offer provably better accuracy and protection against information leakage comparing to the popular 2D-Tree structure, while having only slightly longer searching paths. A detailed discussion and future research directions are deferred to Appendix.

1.2 Other Related Work

Demertzis et al. [12, 13] introduced SRC-search in the context of SSE for range queries, together with the Tree-based DAG (TDAG), which augments a binary tree over the domain with one overlapping intermediate node between every two neighboring canonical ranges at each level, and bounds the size of the returned range in the worst case. Falzon et al. [15] subsequently formalized SRC-search as a general algorithm for range-supporting data structures and extended it to multi-attribute range queries; their framework accommodates data-dependent structures such as KD-Trees, but favors data-independent structures in its concrete constructions, to avoid leakage of the underlying data distribution. Exact covering rules, such as Best and Uniform Range Cover, return several nodes and no false positives [12, 13, 15]; Faber et al. [14] proposed related tree-covering constructions for range, substring, wildcard, and phrase queries, and Kerman-shahi et al. [20] give a further multidimensional instantiation of encrypted range search. Our baseline, the 2D-Tree, is the two-dimensional instance of Bentley’s KD-tree [3, 4]; its probabilistic analysis under random inputs is classical [8], and surveys of geometric range searching appear in [1, 2].

The one-dimensional case of this comparison, SRC-search on c -DAGs against the 1D-Tree, including its average-case analysis and its extensions to non-uniform data, is treated in our companion work [21]. We continue this line of work in three directions: (1) whereas the TDAG and the concrete constructions of [15] partition the domain into subdivisions fixed in advance, our c -2D-DAGs are data-dependent, with canonical ranges computed from the realized elements of the dataset; (2) the TDAG is the case $c = 3$ of our tunable family $c = 2^\alpha + 1$, $\alpha \geq 1$, of overlapping DAGs built on the 2D-Tree; and (3) whereas prior work bounds false positives in the worst case, we carry out an average-case competitive analysis over streams of random range queries, extend it to arbitrary datasets via a learned uniform baseline, and analyze the entropy of the returned node under a randomized choice among all minimal nodes.

2 Preliminaries and Query Stream Model

We formalize the two-dimensional range search model underlying SRC-search on range-supporting data structures. Our goal is to compare SRC-search on the 2D-Tree and on the richer, overlapping c -2D-DAG structures, both defined in Sec. 2.1. The comparison metrics – Returned Level Overhead and Accuracy Ratio – are defined in Section 2.2. Table 1 summarizes the notation.

We consider a K -attribute database $\mathcal{DB}$, $K \geq 2$, following the multi-attribute model of [15]. We restrict our analysis to range queries over two of these attributes, indexed by $i \in \{0, 1\}$, which

we call the range search dimensions; the remaining attributes are carried by every record but are not searched on. We use attribute and dimension interchangeably. For $i \in \{0, 1\}$, we write $[M_i] = \{0, 1, \dots, M_i - 1\}$, with $M_i \geq 2$, for the domain of range search dimension i , i.e., the M_i distinct coordinate positions it may take, so that the two-dimensional domain is $[M_0] \times [M_1]$. Let $D \subseteq [M_0] \times [M_1]$ denote the projection of $\mathcal{DB}$ onto the two range search dimensions, taken without repetition: records of $\mathcal{DB}$ that agree on both range search dimensions are represented by the single domain position they share, and we ignore such multiplicities from here on. We write $N = |D|$ for the number of distinct domain positions realized by $\mathcal{DB}$, and, for $i \in \{0, 1\}$, N_i for the number of distinct coordinates realized by D along dimension i , so that $N_i \leq M_i$ and $N \leq N_0 \cdot N_1 \leq M_0 \cdot M_1$. All structures in this paper are built on D , and all queries are evaluated against D . In Section 3, we assume $D = [M_0] \times [M_1]$, i.e., every domain position holds exactly one element of D (uniform occupancy), so that $N_i = M_i$ and $N = M_0 \cdot M_1$, to obtain closed-form results. Section 4 drops this assumption and extends these results to arbitrary, possibly heterogeneously distributed datasets over the domain, with $N \leq M_0 \cdot M_1$.

A range query is an axis-aligned half-open rectangle $Q = Q^{(0)} \times Q^{(1)} \subseteq [0, M_0) \times [0, M_1)$ with $Q^{(i)} = [x_i, x_i + s_i]$, where $\mathbf{s} = (s_0, s_1)$, with $0 < s_i < M_i$, is the query shape and $\mathbf{x} = (x_0, x_1)$ is the query starting position, satisfying $x_i \in [0, M_i - s_i]$. Starting positions and side lengths are real-valued: we embed the domain in the continuous box $[0, M_0) \times [0, M_1)$, and the elements of D matching Q are those in $D \cap Q$. For $i \in \{0, 1\}$ and a fixed s_i , let $\mathcal{I}_{s_i}$ denote the uniform distribution over the valid starting positions $x_i \in [0, M_i - s_i]$. For a fixed shape $\mathbf{s}$, let $\mathcal{I}_{\mathbf{s}} = \mathcal{I}_{s_0} \times \mathcal{I}_{s_1}$ denote the uniform distribution over all valid starting positions $\mathbf{x} \in [0, M_0 - s_0) \times [0, M_1 - s_1)$; in particular, under $\mathcal{I}_{\mathbf{s}}$ the coordinates x_0 and x_1 are independent. Both structures depend only on the order of the coordinates, thus the coordinates may be real-valued.

To capture a server handling live query traffic, we model queries as an incoming stream $Q_1, Q_2, \dots, Q_\tau$, with $\tau \geq 1$ queries, each drawn independently from $\mathcal{I}_{\mathbf{s}}$ for a fixed shape $\mathbf{s}$, unless stated otherwise. We write $Q_t = Q_t^{(0)} \times Q_t^{(1)}$ for the t -th query of the stream. In more general setting, we consider a stream that may mix $\sigma \geq 1$ distinct shapes $\mathbf{s}_j = (s_j^{(0)}, s_j^{(1)})$, $j \in \{1, \dots, \sigma\}$, with frequencies $\omega_1, \dots, \omega_\sigma > 0$, $\sum_{j=1}^\sigma \omega_j = 1$: each query independently selects shape $\mathbf{s}_j$ with probability ω_j and then draws its starting position from $\mathcal{I}_{\mathbf{s}_j}$.

2.1 Range-Supporting Structures and Covering Algorithms

Following [15], a range-supporting data structure over $[0, M_0) \times [0, M_1)$ is a pair $(\mathcal{G}, \text{RC})$. Here $\mathcal{G} = (V, E)$ is a connected directed acyclic graph with a single source node, the root, in which every node v stores a canonical range $\text{range}_v \subseteq [0, M_0) \times [0, M_1)$, with $\text{range}_{\text{root}} = [0, M_0) \times [0, M_1)$ and $\text{range}_v = \bigcup_{(v,w) \in E} \text{range}_w$ for every non-leaf node v . In the structures we study, every canonical range is an axis-aligned rectangle $\text{range}_v = \text{range}_v^{(0)} \times \text{range}_v^{(1)}$ whose sides $\text{range}_v^{(i)} \subseteq [0, M_i)$ are half-open intervals. Each node v is associated with the elements $D_v = D \cap \text{range}_v$; in particular, $D_{\text{root}} = D$ and $|D_{\text{root}}| = N$.

The component RC is a range covering rule: it maps a query Q to a set $W \subseteq V$ of nodes, called a cover of Q , such that $Q \subseteq \bigcup_{w \in W} \text{range}_w$. The elements of $(\bigcup_{w \in W} D_w) \setminus Q$ are the false positives of W : they are returned together with the answer $D \cap Q$ and filtered out client-side.

Exact covering rules, such as Best Range Cover (BRC) and Uniform Range Cover (URC) [12, 13, 15], avoid false positives at the cost of covers that in general consist of several nodes; they are not the object of this paper. We study the single range cover (SRC) rule, introduced in [12, 13] and formalized for general range-supporting structures in [15], which covers Q with a single node. In [15], SRC returns a node v with $Q \subseteq \text{range}_v$ that minimizes the number of domain positions of range_v outside Q . Since the structures below are data-dependent and false positives are elements of D , we minimize the element count instead: SRC-search on $\mathcal{G}$ returns a node

$$v_{\mathcal{G}}(Q) \in \arg \min \{ |D_v| : v \in V, Q \subseteq \text{range}_v \},$$

whose $|D_{v_{\mathcal{G}}(Q)}| - |D \cap Q|$ false positives are the elements of $D_{v_{\mathcal{G}}(Q)} \setminus Q$. The set on the right-hand side is nonempty, since the root contains every query. Under the uniform occupancy assumption of Section 3, the two objectives, minimizing $|D_v|$ and minimizing the number of domain positions of range_v outside Q , coincide. If several nodes attain the minimum, SRC-search returns one of them, a deepest one if they lie at different levels; this choice affects neither $|D_{v_{\mathcal{G}}(Q)}|$ nor $AR(Q)$, and under uniform occupancy all minimizers lie at the same level. The exhaustive SRC-search of [15], adapted to this objective, descends from the root into every child whose canonical range contains Q and returns a visited node minimizing $|D_v|$.

The 2D-Tree and the c -2D-DAG defined below are data-dependent: their canonical ranges are not fixed subdivisions of the domain chosen in advance, but are computed from the elements of D ; we assume $N \geq 1$. Both structures are leveled: the root forms level 0, every edge leads from a node at some level ℓ to a node at level $\ell + 1$, and we write $\text{level}(v)$ for the level of node v . In both structures, a node v is a leaf if $|D_v| \leq 1$; otherwise v is refined along a split dimension $i(v) \in \{0, 1\}$, defined as follows for v at level ℓ : $i(v) = \ell \bmod 2$ if D_v realizes at least two distinct coordinates along dimension $\ell \bmod 2$, and $i(v) = 1 - (\ell \bmod 2)$ otherwise. Since the elements of D_v are pairwise distinct and $|D_v| \geq 2$, D_v realizes at least two distinct coordinates along dimension $i(v)$.

For a non-leaf node v , let $u_1 < \dots < u_r$, with $r \geq 2$, be the distinct coordinates realized by D_v along dimension $i(v)$. Every child w of v defined below spans a set of consecutive ranks $\{g, \dots, h\}$, with $1 \leq g \leq h \leq r$: range_w keeps the side of range_v along dimension $1 - i(v)$, and its side along dimension $i(v)$ is $[u_g, u_{h+1})$, where the left endpoint is replaced by that of $\text{range}_v^{(i(v))}$ if $g = 1$, and the right endpoint by that of $\text{range}_v^{(i(v))}$ if $h = r$. Hence D_w consists of the elements of D_v whose $i(v)$ -th coordinate lies in $\{u_g, \dots, u_h\}$, and $|D_w| \geq 1$.

The 2D-Tree. The baseline structure is the 2D-Tree over D , the two-dimensional instance of Bentley's KD-Tree [3, 5]. Each non-leaf node v has two children: the child spanning ranks 1 to $\lceil r/2 \rceil$ and the other child spanning ranks $\lceil r/2 \rceil + 1$ to r . Their canonical ranges partition range_v , and their element sets partition D_v into two nonempty sets. Consequently, the 2D-Tree has N leaves, each holding one element, and $2N - 1$ nodes. A split along dimension i maps the number of distinct coordinates realized along dimension i from r to $\lceil r/2 \rceil$ or $\lfloor r/2 \rfloor$, and a split along the other dimension does not increase it; hence every root-to-leaf path contains at most $\lceil \log_2 N \rceil$ splits along dimension i , and the height of the 2D-Tree is at least $\lceil \log_2 N \rceil$ and at most $\lceil \log_2 N_0 \rceil + \lceil \log_2 N_1 \rceil$. Because the children of every node partition its canonical range, the canonical ranges at any fixed level

are pairwise disjoint, and the nodes whose canonical ranges contain a query Q form a path starting at the root. Along this path, $|D_v|$ strictly decreases, so $v_{\text{Tree}}(Q)$ is the deepest node of the path and is the unique minimizer in the definition of SRC-search. We write $\text{level}_{\text{Tree}}(Q) = \text{level}(v_{\text{Tree}}(Q))$.

The c-2D-DAG Family. We compare the 2D-Tree with a family of overlapping, data-dependent DAGs, called c -2D-DAGs, built on top of it. The branching parameter $c = 2^\alpha + 1$, for an integer $\alpha \geq 1$, controls the density of canonical ranges. Every non-leaf node v has children $w_1, \dots, w_c$, each spanning $\lceil r/2 \rceil$ or $\lfloor r/2 \rfloor$ consecutive ranks. The first and last children, w_1 and w_c , are the two 2D-Tree children of v defined above. The remaining $c - 2$ augmented children $w_2, \dots, w_{c-1}$ are the children spanning $\lceil r/2 \rceil$ ranks starting at rank $j \cdot \lfloor r/(2(c-1)) \rfloor + 1$, for $j = 1, \dots, c-2$; that is, w_{j+1} spans ranks $j \cdot \lfloor r/(2(c-1)) \rfloor + 1$ to $j \cdot \lfloor r/(2(c-1)) \rfloor + \lceil r/2 \rceil$. One checks that $(c-2) \cdot \lfloor r/(2(c-1)) \rfloor + \lceil r/2 \rceil \leq r$, so every augmented child spans ranks within $\{1, \dots, r\}$. Since w_1 and w_c already partition range_v , the union of the canonical ranges of v 's children is exactly range_v . If $\lfloor r/(2(c-1)) \rfloor \geq 1$, i.e., $r \geq 2(c-1)$, the c children are pairwise distinct and consecutive children overlap along dimension $i(v)$; if $\lfloor r/(2(c-1)) \rfloor = 0$, all augmented children coincide with w_1 . For $c = 3$, the single augmented child w_2 starts at rank $\lfloor r/4 \rfloor + 1$ and straddles the boundary between w_1 and w_3 ; for $c = 5$, the augmented children w_2, w_3, w_4 start at ranks $\lfloor r/8 \rfloor + 1$, $2\lfloor r/8 \rfloor + 1$, and $3\lfloor r/8 \rfloor + 1$. For $\alpha = 0$, i.e., $c = 2$, only the two 2D-Tree children remain, and the construction yields the 2D-Tree.

The c -2D-DAG is built level by level: level 0 consists of the root, and level $\ell + 1$ consists of the distinct rectangles that occur as children of level ℓ nodes, where identical rectangles generated at the same level form a single node, linked to each of its parents. Since the split dimension and the children of a node depend only on its level, its canonical range, and its element set, this identification is consistent. Every edge connects consecutive levels, so the c -2D-DAG is a leveled DAG with the root as its single source, and, together with SRC-search, it is a range-supporting data structure in the sense defined above. The 2D-Tree is embedded in every c -2D-DAG: by induction on ℓ , every level ℓ node of the 2D-Tree is a level ℓ node of the c -2D-DAG with the same canonical range and element set, hence with the same split dimension, and its two 2D-Tree children are its children w_1 and w_c in the c -2D-DAG. Consequently, SRC-search on the c -2D-DAG minimizes $|D_v|$ over a superset of the candidates available in the 2D-Tree, and $|D_{v_{c\text{-DAG}}(Q)}| \leq |D_{v_{\text{Tree}}(Q)}|$ for every query Q .

Because children overlap, several nodes of the same level may contain Q , so, unlike on the 2D-Tree, SRC-search on the c -2D-DAG does not follow a single root-to-node path, and the node it returns need not be unique. Since the parents of a node containing Q also contain Q , the exhaustive SRC-search of [15], which descends into every child containing Q , examines every node containing Q . We write $\text{level}_{c\text{-DAG}}(Q) = \text{level}(v_{c\text{-DAG}}(Q))$. Under uniform occupancy and dyadic assumptions of Section 3, $|D_v|$ depends only on $\text{level}(v)$ and strictly decreases with it, so SRC-search on either structure returns a deepest node containing Q , and the embedding above yields $\text{level}_{\text{Tree}}(Q) \leq \text{level}_{c\text{-DAG}}(Q)$; this is the comparison formalized by the Returned Level Overhead in Section 2.2. Fig. 2 illustrates both structures for $c = 3$ over a uniformly occupied 16×16 domain. For Q_1 , the augmented children allow the 3-2D-DAG to return a node at level 4 with 16 elements: both $[2, 6) \times [0, 4)$ and $[2, 6) \times [2, 6)$ contain Q_1 and attain the minimum, so SRC-search returns one of them, while

the 2D-Tree, whose canonical ranges are disjoint at each level, returns the node $[0, 8) \times [0, 8)$ at level 2 with 64 elements; for Q_2 , both structures return the same node.

2.2 Performance Measures

Consider two servers indexing the same dataset D in parallel, one using the 2D-Tree and the other the c -2D-DAG, for some $c = 2^\alpha + 1$ with $\alpha \geq 1$, and let every query Q_t of the stream be answered by both. We compare the two structures by two measures: an additive Returned Level Overhead and a multiplicative Accuracy Ratio. A larger c provides a denser family of overlapping canonical ranges, which may improve accuracy at the cost of a larger structure and a more expensive search. We first define both measures for a single query and then average them over a stream of queries.

Returned Level (Overhead). For $\mathcal{G}$ the 2D-Tree or the c -2D-DAG and a query Q , the returned node $v_{\mathcal{G}}(Q)$ and the returned level $\text{level}_{\mathcal{G}}(Q) = \text{level}(v_{\mathcal{G}}(Q))$ are defined in Section 2.1. We use the returned level, rather than wall-clock search time, as our cost measure because running time is implementation-dependent. On the 2D-Tree, SRC-search follows a single, forced path. On the c -2D-DAG, several children of a node may contain Q , and the exhaustive SRC-search of [15] descends into each of them to find a node minimizing $|D_v|$, at a higher per-level cost than on the 2D-Tree. The returned level abstracts from the implementation costs and isolates the structural depth analyzed in the rest of the paper. For a single query Q , the Returned Level Overhead of the c -2D-DAG relative to the 2D-Tree is

$$\text{level}_{\text{diff}}(Q) = \text{level}_{c\text{-DAG}}(Q) - \text{level}_{\text{Tree}}(Q),$$

the number of additional levels the c -2D-DAG descends to return its node. Under uniform occupancy and dyadic assumptions of Section 3, $\text{level}_{\text{diff}}(Q) \geq 0$ for every query Q (Sections 2.1 and 3).

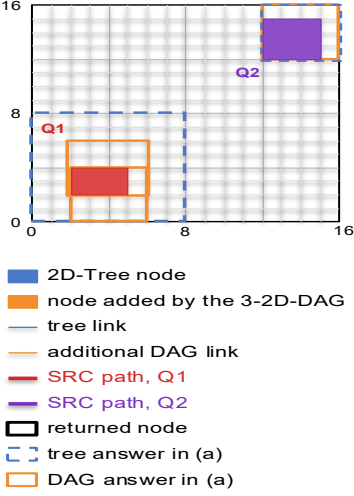
Accuracy (Ratio). For $\mathcal{G}$ the 2D-Tree or the c -2D-DAG and a query Q , the returned node $v_{\mathcal{G}}(Q)$ is associated with the elements $D_{v_{\mathcal{G}}(Q)}$. If $D \cap Q \neq \emptyset$, the accuracy of the returned node is $A_{\mathcal{G}}(Q) = |D_{v_{\mathcal{G}}(Q)}|/|D \cap Q|$. Since $Q \subseteq \text{range}_{v_{\mathcal{G}}(Q)}$ implies $D \cap Q \subseteq D_{v_{\mathcal{G}}(Q)}$, we have $A_{\mathcal{G}}(Q) \geq 1$, with equality exactly when the returned node contains no false positives; larger values indicate more false positives, i.e., lower accuracy. For a single query Q , the Accuracy Ratio of the 2D-Tree relative to the c -2D-DAG is

$$AR(Q) = \frac{A_{\text{Tree}}(Q)}{A_{c\text{-DAG}}(Q)} = \frac{|D_{v_{\text{Tree}}(Q)}|}{|D_{v_{c\text{-DAG}}(Q)}|},$$

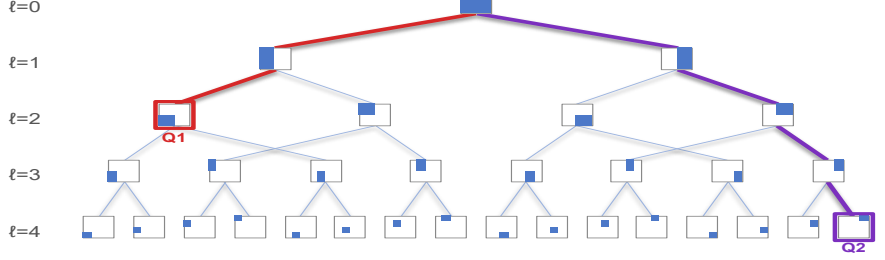
which is defined for every query, since every node holds at least one element (Section 2.1), and equals $A_{\text{Tree}}(Q)/A_{c\text{-DAG}}(Q)$ whenever $D \cap Q \neq \emptyset$. By the embedding of the 2D-Tree in the c -2D-DAG (Section 2.1), $AR(Q) \geq 1$ for every dataset and every query. The Accuracy Ratio is the factor by which the number of elements returned by the 2D-Tree exceeds the number returned by the c -2D-DAG for the same query; a larger ratio indicates tighter containment by the c -2D-DAG. Under uniform occupancy, every canonical range has integer endpoints, so $|D_v|$ equals the area of range_v , and $AR(Q)$ is the ratio of the areas of the two returned rectangles; under the additional dyadic assumption of Section 3, $AR(Q) = 2^{\text{level}_{\text{diff}}(Q)}$.

Averaging over a Stream of Queries. Given a stream of queries $Q_1, \dots, Q_r$, we average the two per-query measures over the

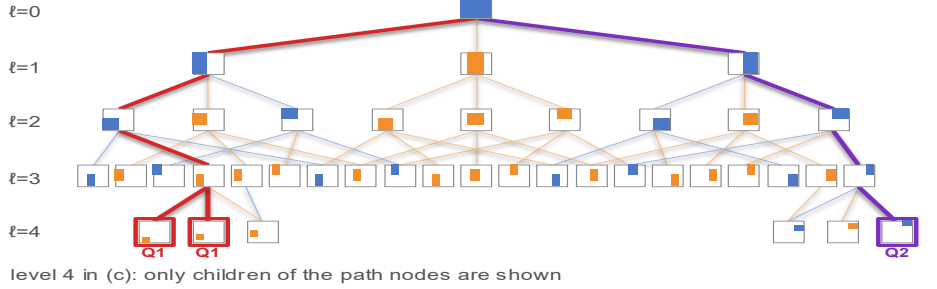
(a) Domain, queries, returned ranges



(b) 2D-Tree



(c) 3-2D-DAG



level 4 in (c): only children of the path nodes are shown

Figure 2: The 2D-Tree and the 3-2D-DAG over the uniformly occupied domain $D = [16] \times [16]$, for two queries: $Q_1 = [2, 5] \times [2, 4]$ (red, shape (3, 2)) and $Q_2 = [12, 15] \times [12, 15]$ (purple, shape (3, 3)). (a) The queries and the canonical ranges returned by SRC-search: dashed blue for the 2D-Tree, solid orange for the 3-2D-DAG (two nodes for Q_1 , which tie). (b), (c) Levels 0 to 4 of the two structures; each square depicts the canonical range of a node inside the domain. Blue nodes and links belong to the 2D-Tree. Orange nodes and links are added by the 3-2D-DAG. Bold edges show a root-to-node path to the returned node (boxed). For Q_1 , the 2D-Tree returns a level 2 node with 64 elements, whereas two level 4 nodes of the 3-2D-DAG contain Q_1 , both with 16 elements, and SRC-search returns either of them (both boxed); thus $\text{level}_{\text{diff}}(Q_1) = 2$ and $\text{AR}(Q_1) = 4$. For Q_2 , both structures return the same level 4 node and $\text{level}_{\text{diff}}(Q_2) = 0$. In (c), level 4 shows only the children of the path nodes.

stream. The average Returned Level Overhead is

$$\overline{\text{level}_{\text{diff}}} = \frac{1}{\tau} \sum_{t=1}^{\tau} \text{level}_{\text{diff}}(Q_t), \quad (1)$$

and the average Accuracy Ratio is

$$\overline{\text{AR}} = \frac{1}{\tau} \sum_{t=1}^{\tau} \text{AR}(Q_t). \quad (2)$$

Both are averages over the given query sequence; no distributional assumption on $Q_1, \dots, Q_\tau$ is required to define them. Under the assumptions of Section 3, $\overline{\text{AR}} = \frac{1}{\tau} \sum_{t=1}^{\tau} 2^{\text{level}_{\text{diff}}(Q_t)}$.

3 Theoretical Analysis

Assumptions. Throughout this section, we work under the following two assumptions on the dataset:

- Uniform occupancy: $D = [M_0] \times [M_1]$, i.e., every domain position holds exactly one element of D ; hence $N_i = M_i$ and $N = N_0 N_1$.
- Dyadic domain: $N_i = 2^{n_i+1}$ for integers $n_i \geq 0$, $i \in \{0, 1\}$.

Lemma 3.1 collects the consequences of uniform occupancy and dyadic domain that we use freely in the rest of this section: both structures become data-independent, the element count of every node equals the area of its canonical range and depends only on its level, and, at the levels relevant to SRC-search, the splits alternate strictly between the two dimensions. These assumptions are relaxed in Section 4.

LEMMA 3.1 (STRUCTURE UNDER UNIFORM OCCUPANCY AND DYADIC DOMAIN). Assume uniform occupancy and dyadic domain, and let $\mathcal{G}$ be the c-2D-DAG for $c = 2^\alpha + 1$ with $\alpha \geq 0$, where $\alpha = 0$ yields the 2D-Tree. Then:

- Every canonical range is a product of two half-open intervals with integer endpoints whose lengths are powers of 2, D_v consists of all integer points of range_v , and $|D_v|$ equals the area of range_v .
- Every node v at level ℓ satisfies $|D_v| = N/2^\ell$. The leaves of $\mathcal{G}$ are exactly its nodes at level $n_0 + n_1 + 2 = \log_2(N)$, which is the height of $\mathcal{G}$.
- Let $\hat{\ell} = \min(2n_0 + 2, 2n_1 + 3)$. For $0 \leq \ell \leq \hat{\ell}$, every node at level ℓ has side lengths $N_0/2^{\lceil \ell/2 \rceil}$ along dimension 0 and $N_1/2^{\lfloor \ell/2 \rfloor}$ along dimension 1, and for $0 \leq \ell < \hat{\ell}$, every node at level ℓ has split dimension $\ell \bmod 2$.
- Let v be a non-leaf node whose side along $i(v)$ is $[a, a + L)$. If $L \geq 2(c - 1)$, then for $j = 0, \dots, c - 1$ the side of w_{j+1} along $i(v)$ is $[a + j \frac{L}{2(c-1)}, a + j \frac{L}{2(c-1)} + \frac{L}{2})$; in particular, w_1 and w_c split range_v at its geometric midpoint $a + L/2$. If $L < 2(c - 1)$, the augmented children of v coincide with w_1 .

Query shapes. For $i \in \{0, 1\}$ and a side length s_i , let

$$\kappa_i = \lfloor \log_2(N_i/s_i) \rfloor, \quad \text{equivalently} \quad \frac{N_i}{2^{\kappa_i+1}} < s_i \leq \frac{N_i}{2^{\kappa_i}},$$

so that κ_i is the largest number of halvings of $[0, N_i]$ after which an interval can still contain $Q^{(i)}$. Throughout this section, we assume that the query shape $\mathbf{s} = (s_0, s_1)$ satisfies the following shape assumptions: for $i \in \{0, 1\}$, $1 < s_i < N_i$ and $2^{\kappa_i} \leq N_i/(c -$

Table 1: Notation used in the paper (symbols used only inside proofs are omitted).

Symbol	Meaning
<u>Data and domain</u>	
$\mathcal{DB}, K$	K -attribute database, $K \geq 2$
$i \in \{0, 1\}$	index of a range search dimension
$[M_i], M_i$	domain $\{0, \dots, M_i - 1\}$ of dimension i ; its size, $M_i \geq 2$
D	projection of $\mathcal{DB}$ onto the two range search dimensions, without repetition, $D \subseteq [M_0] \times [M_1]$
N, N_i	$N = D $; number of distinct coordinates realized by D along dimension i
n_i	dyadic exponent, $N_i = 2^{n_i+1}$, $n_i \geq 0$
<u>Queries and streams</u>	
$Q = Q^{(0)} \times Q^{(1)}$	range query (axis-aligned half-open rectangle), $Q^{(i)} = [x_i, x_i + s_i)$
$\mathbf{s} = (s_0, s_1)$	query shape, $0 < s_i < M_i$
$\mathbf{x} = (x_0, x_1)$	query starting position, $x_i \in [0, M_i - s_i]$
$\mathcal{I}_s, \mathcal{I}_s$	uniform distribution over valid x_i ; over valid $\mathbf{x}$, $\mathcal{I}_s = \mathcal{I}_{s_0} \times \mathcal{I}_{s_1}$
$Q_1, \dots, Q_\tau; \tau$	stream of queries; its length, $\tau \geq 1$
$Q_t = Q_t^{(0)} \times Q_t^{(1)}$	t -th query of the stream
$\sigma; s_j = (s_j^{(0)}, s_j^{(1)})$	number of distinct shapes in a mixed stream; j -th shape, $j \in \{1, \dots, \sigma\}$
ω_j	frequency of shape s_j , $\omega_j > 0$, $\sum_{j=1}^\sigma \omega_j = 1$
<u>Range-supporting structures</u>	
$(\mathcal{G}, \text{RC})$	range-supporting data structure; RC is its range covering rule
$\mathcal{G} = (V, E)$	connected DAG with a single source node, the root
$\text{range}_v, \text{range}_v^{(i)}$	canonical range of node v ; its side along dimension i
D_v	elements of D in node v : $D_v = D \cap \text{range}_v$; its size is $ D_v $
W	cover of Q : $W \subseteq V$ with $Q \subseteq \bigcup_{w \in W} \text{range}_w$
$\text{level}(v)$	level of node v ; the root forms level 0
$i(v)$	split dimension of a non-leaf node v
$u_1 < \dots < u_r$	distinct coordinates realized by D_v along $i(v)$, $r \geq 2$
$c = 2^\alpha + 1$	branching parameter of the c -2D-DAG, integer $\alpha \geq 1$
$w_1, \dots, w_c$	children of a non-leaf node v of the c -2D-DAG; w_1, w_c are its 2D-Tree children, $w_2, \dots, w_{c-1}$ its augmented children

Symbol	Meaning
<u>SRC-search and measures</u> ($\mathcal{G}$ is the 2D-Tree or the c -2D-DAG)	
$v_{\mathcal{G}}(Q)$	node returned by SRC-search on $\mathcal{G}$: $\arg \min\{ D_v : v \in V, Q \subseteq \text{range}_v\}$
$\text{level}_{\mathcal{G}}(Q)$	returned level, $\text{level}(v_{\mathcal{G}}(Q))$; written $\text{level}_{\text{Tree}}(Q)$, $\text{level}_{c\text{-DAG}}(Q)$
$A_{\mathcal{G}}(Q)$	accuracy: $ D_{v_{\mathcal{G}}(Q)} / D \cap Q \geq 1$
$\text{level}_{\text{diff}}(Q)$	Returned Level Overhead, $\text{level}_{c\text{-DAG}}(Q) - \text{level}_{\text{Tree}}(Q)$
$AR(Q)$	Accuracy Ratio: $A_{\text{Tree}}(Q)/A_{c\text{-DAG}}(Q) = D_{v_{\text{Tree}}(Q)} / D_{v_{c\text{-DAG}}(Q)} $
$\overline{\text{level}_{\text{diff}}}, \overline{AR}$	average Returned Level Overhead and average Accuracy Ratio over $Q_1, \dots, Q_\tau$
<u>Analysis</u>	
$\Pr_{\mathcal{I}_s}[\cdot], \mathbb{E}[\cdot]$	probability for Q drawn from $\mathcal{I}_s$; expectation
$\hat{\ell}$	$\min(2n_0 + 2, 2n_1 + 3)$; every level $\ell < \hat{\ell}$ has split dimension $\ell \bmod 2$
κ_i	$\lfloor \log_2(N_i/s_i) \rfloor$
L^*	$\min(2\kappa_0, 2\kappa_1 + 1)$; no node at a level greater than L^* contains a query of shape $\mathbf{s}$
d	depth of a dyadic or canonical interval, integer $d \geq 0$
$\lambda_i(Q)$	largest d such that $Q^{(i)}$ is contained in a depth d dyadic interval of dimension i
$\mu_i(Q)$	largest d such that $Q^{(i)}$ is contained in a depth d canonical interval of dimension i
p_0, p_1	$\Pr_{\mathcal{I}_s}[\mu_0(Q) \geq \lceil L^*/2 \rceil], \Pr_{\mathcal{I}_s}[\mu_1(Q) \geq \lfloor L^*/2 \rfloor]$
η_0, η_1	$\max\{0, (c-1)s_i - (c-2)N_i/2^{\kappa_i}\}$ under the case conditions of the frame defining TQD, and 0 otherwise
$\kappa_j^{(i)}, m_j$	$\lfloor \log_2(N_i/s_j^{(i)}) \rfloor$; $m_j = \min\{\kappa_j^{(0)}, \kappa_j^{(1)}\}$
$\bar{m}$	$\sum_{j=1}^\sigma \omega_j m_j$
Γ	$\prod_{j=1}^\sigma \min\{N_0/s_j^{(0)}, N_1/s_j^{(1)}\}^{\omega_j}$, the ω -weighted geometric mean of $\min\{N_0/s_j^{(0)}, N_1/s_j^{(1)}\}$
β	level-balance factor of D (Section 4)
N^*, s^*, π_k^*	dyadic size, shape, and pmf of a baseline LOD $\text{TQD}_c(N^*, s^*)$
ε	L_2 distance between the LOD of D and a baseline LOD

1). The first condition gives $\kappa_i \leq n_i$. Since $c - 1 = 2^\alpha$, the second condition is equivalent to $\kappa_i \leq n_i + 1 - \alpha$; for $c = 3$ it therefore follows from the first. Let

$$L^* = \min(2\kappa_0, 2\kappa_1 + 1).$$

Since $\kappa_i \leq n_i$, we have $L^* + 1 \leq \min(2n_0 + 1, 2n_1 + 2) \leq \hat{\ell}$, so Lemma 3.1(iii) applies to every level $\ell \leq L^* + 1$. By this lemma, the nodes at level ℓ have both sides at least as long as the corresponding sides of $\mathbf{s}$ if and only if $\ell \leq L^*$; in particular, no node at level $L^* + 1$ contains a query of shape $\mathbf{s}$. Every node at a deeper level is contained in an ancestor at level $L^* + 1$. Hence no node at a level greater than L^* contains a query of shape $\mathbf{s}$. Queries with $s_i \leq 1$ for some i may be contained in nodes at levels beyond $\hat{\ell}$, where the splits no longer alternate, and are not covered by the results below. By the definition of L^* , every node at a level $\ell < L^*$ has been halved at most $\kappa_i - 1$ times along its split dimension i , therefore by the second condition its side length there is at least $2N_i/2^{\kappa_i} \geq 2(c-1)$. Hence, by Lemma 3.1(iv), its c children are pairwise distinct and evenly spaced.

Section 2.2 introduced two per-query comparative measures, the Returned Level Overhead $\text{level}_{\text{diff}}(Q)$ and the Accuracy Ratio $AR(Q)$, together with their stream averages $\overline{\text{level}_{\text{diff}}}$ and $\overline{AR}$. This section derives these four quantities under the above assumptions. By Lemma 3.1(ii), SRC-search on either structure returns a deepest node containing Q , so $\text{level}_{\text{diff}}(Q) \geq 0$ by Section 2.1, and $AR(Q) = 2^{\text{level}_{\text{diff}}(Q)}$ for every query Q . Section 3.1 first characterizes the level returned by SRC-search on the 2D-Tree and on the

c -2D-DAG separately (Lemmas 3.2 and 3.3), and then combines the two into a technical step, the Level Overhead Distribution (LOD), the probability mass function of $\text{level}_{\text{diff}}(Q)$ for a query drawn from $\mathcal{I}_s$ (Lemma 3.4). Section 3.2 then lifts this single-query result to a stream of queries, bounding $\overline{\text{level}_{\text{diff}}}$ and $\overline{AR}$ for streams of a single query shape (Theorem 3.5) and, more generally, of mixed query shapes (Corollary 3.6).

3.1 2D-Tree vs. c -2D-DAG: Level Overhead Distribution (LOD)

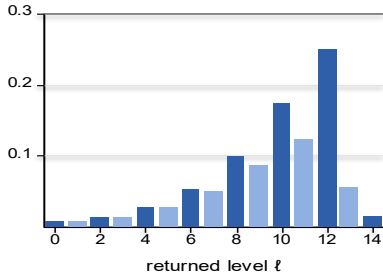
This section derives the Level Overhead Distribution (LOD) in two steps. Lemma 3.2 gives the level distribution of SRC-search on the 2D-Tree, Lemma 3.3 the analogous distribution for the c -2D-DAG, and Lemma 3.4 combines them into the distribution of $\text{level}_{\text{diff}}(Q)$ for a query Q drawn from $\mathcal{I}_s$. Both lemmas reduce the two-dimensional returned level to two per-dimension containment depths, which depend only on x_0 and on x_1 , respectively, and are therefore independent under $\mathcal{I}_s$. For the 2D-Tree, these depths are defined as follows. For $i \in \{0, 1\}$ and an integer $d \geq 0$, the depth- d dyadic intervals of dimension i are the 2^d intervals $[jN_i/2^d, (j+1)N_i/2^d]$, $j \in \{0, \dots, 2^d - 1\}$, and

$$\lambda_i(Q) = \max\{d \geq 0 : Q^{(i)} \text{ is contained in a depth-}d \text{ dyadic interval of dimension } i\}.$$

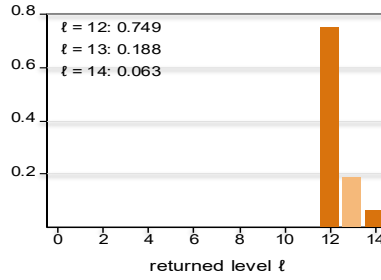
Since $Q^{(i)} \subseteq [0, N_i]$ and a depth- d dyadic interval can contain $Q^{(i)}$ only if $N_i/2^d \geq s_i$, we have $0 \leq \lambda_i(Q) \leq \kappa_i$.

$N_0 = N_1 = 1024$, $s_0 = s_1 = 7$, $c = 3$ ($\kappa_0 = \kappa_1 = 7$, $L^* = 14$)

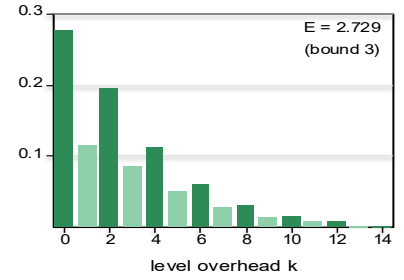
(a) 2D-Tree



(b) 3-2D-DAG



(c) Level overhead (LOD)



Even and odd levels are shaded differently (dark: even, light: odd). Probabilities are exact, from Lemmas 3.2, 3.3 and 3.4.

Figure 3: Exact level distributions for $N_0 = N_1 = 1024$, $s_0 = s_1 = 7$ and $c = 3$ ($\kappa_0 = \kappa_1 = 7$, $L^* = 14$), computed from Lemmas 3.2, 3.3 and 3.4, respectively. (a) Returned level of the 2D-Tree. (b) Returned level of the 3-2D-DAG, supported on $\{12, 13, 14\}$. (c) Level overhead level_{diff}; its mean is 2.729. Note the different vertical scales.

LEMMA 3.2 (LEVEL DISTRIBUTION OF SRC-SEARCH ON THE 2D-TREE). Assume uniform occupancy, dyadic domain, and $1 < s_i < N_i$ for $i \in \{0, 1\}$, and let $Q = Q^{(0)} \times Q^{(1)}$ be a random query drawn from $\mathcal{I}_s$. Then:

- (i) $\text{level}_{\text{Tree}}(Q) = \min(2\lambda_0(Q), 2\lambda_1(Q) + 1)$, where $\lambda_0(Q)$ and $\lambda_1(Q)$ are independent and, for $i \in \{0, 1\}$ and $0 \leq d \leq \kappa_i$,

$$\Pr_{\mathcal{I}_s}[\lambda_i(Q) \geq d] = \frac{N_i - 2^d s_i}{N_i - s_i}. \quad (3)$$

- (ii) $\text{level}_{\text{Tree}}(Q) \in \{0, \dots, L^*\}$ and $\Pr_{\mathcal{I}_s}[\text{level}_{\text{Tree}}(Q) = \ell]$ equals

$$\begin{cases} \frac{2^{\ell/2} s_0 (N_1 - 2^{\ell/2} s_1)}{(N_0 - s_0)(N_1 - s_1)}, & \ell \text{ even}, \ell < L^*, \\ \frac{2^{(\ell-1)/2} s_1 (N_0 - 2^{(\ell+1)/2} s_0)}{(N_0 - s_0)(N_1 - s_1)}, & \ell \text{ odd}, \ell < L^*, \\ \frac{(N_0 - 2^{\lfloor L^*/2 \rfloor} s_0)(N_1 - 2^{\lfloor L^*/2 \rfloor} s_1)}{(N_0 - s_0)(N_1 - s_1)}, & \ell = L^*. \end{cases}$$

For the c -2D-DAG, the per-dimension depths are defined analogously. For $i \in \{0, 1\}$ and an integer $d \geq 0$, the depth- d canonical intervals of dimension i are the $(c-1)2^d - (c-2)$ intervals $[jN_i/(2^d(c-1)), jN_i/(2^d(c-1)) + N_i/2^d]$, $j \in \{0, \dots, (c-1)(2^d-1)\}$, and

$$\mu_i(Q) = \max\{d \geq 0 : Q^{(i)} \text{ is contained in a depth-}d \text{ canonical interval of dimension } i\}.$$

Since $Q^{(i)} \subseteq [0, N_i]$ and a depth- d canonical interval can contain $Q^{(i)}$ only if $N_i/2^d \geq s_i$, we have $0 \leq \mu_i(Q) \leq \kappa_i$. Every depth- d dyadic interval is a depth- d canonical interval, obtained for j a multiple of $c-1$; hence $\lambda_i(Q) \leq \mu_i(Q)$. Moreover, the pair $(\lambda_i(Q), \mu_i(Q))$ depends only on x_i , so the pairs $(\lambda_0(Q), \mu_0(Q))$ and $(\lambda_1(Q), \mu_1(Q))$ are independent under $\mathcal{I}_s$.

LEMMA 3.3 (LEVEL DISTRIBUTION OF SRC-SEARCH ON THE c -2D-DAG). Assume uniform occupancy, dyadic domain, and the shape assumptions, let $c = 2^\alpha + 1$ for an integer $\alpha \geq 1$, and let $Q = Q^{(0)} \times Q^{(1)}$ be a random query drawn from $\mathcal{I}_s$. Then:

- (i) $\text{level}_{c\text{-DAG}}(Q) = \min(2\mu_0(Q), 2\mu_1(Q) + 1)$, where $\mu_0(Q)$ and $\mu_1(Q)$ are independent and, for $i \in \{0, 1\}$ and $0 \leq d \leq \kappa_i$, $\Pr_{\mathcal{I}_s}[\mu_i(Q) \geq d]$ equals

$$\begin{cases} 1, & s_i \leq \frac{c-2}{c-1} \cdot \frac{N_i}{2^d}, \\ \frac{((c-1)2^d - (c-2))(N_i/2^d - s_i)}{N_i - s_i}, & \text{otherwise.} \end{cases} \quad (4)$$

- (ii) Let $p_0 = \Pr_{\mathcal{I}_s}[\mu_0(Q) \geq \lceil L^*/2 \rceil]$ and $p_1 = \Pr_{\mathcal{I}_s}[\mu_1(Q) \geq \lfloor L^*/2 \rfloor]$, both given by (4), since $\lceil L^*/2 \rceil \leq \kappa_0$ and $\lfloor L^*/2 \rfloor \leq \kappa_1$. Then $\text{level}_{c\text{-DAG}}(Q) \in \{L^* - 2, L^* - 1, L^*\}$, and

$$\Pr_{\mathcal{I}_s}[\text{level}_{c\text{-DAG}}(Q) = \ell] = \begin{cases} p_0 p_1, & \ell = L^*, \\ p_0 (1 - p_1), & \ell = L^* - 1, L^* \text{ even}, \\ (1 - p_0) p_1, & \ell = L^* - 1, L^* \text{ odd}, \\ 1 - p_0, & \ell = L^* - 2, L^* \text{ even}, \\ 1 - p_1, & \ell = L^* - 2, L^* \text{ odd}. \end{cases}$$

PROOF SKETCH OF LEMMAS 3.2 AND 3.3. Both proofs follow the same three steps. **(1) One dimension at a time.** For $i \in \{0, 1\}$ and $0 \leq d \leq \kappa_i$, $Q^{(i)}$ lies in a given depth- d dyadic (resp. canonical) interval if and only if x_i lies in a window of length $N_i/2^d - s_i$. The 2^d dyadic windows are separated by gaps of length s_i , which gives (3). The canonical windows are shifted by $N_i/(2^d(c-1))$ and separated by signed gaps of length $s_i - \frac{c-2}{c-1} \cdot \frac{N_i}{2^d}$; since $s_i \leq N_i/2^{\kappa_i}$, they overlap or touch for every $d < \kappa_i$, which gives (4) and $\mu_i(Q) \geq \kappa_i - 1$ (cf. Figure 9(a) in the Appendix). Since $\lambda_i(Q)$ and $\mu_i(Q)$ depend only on x_i , the two dimensions are independent. **(2) From depths to levels.** By Lemma 3.1, for $\ell \leq L^*$ the canonical ranges of the level ℓ nodes of either structure are exactly the products of a depth- $\lceil \ell/2 \rceil$ interval of dimension 0 and a depth- $\lfloor \ell/2 \rfloor$ interval of dimension 1 (dyadic for the 2D-Tree, canonical for the c -2D-DAG), and no node at a level greater than L^* contains Q . Hence some level ℓ node contains Q if and only if $\ell \leq 2\lambda_0(Q)$ and $\ell \leq 2\lambda_1(Q) + 1$ (resp. with $\mu_i(Q)$), and, since SRC-search returns a deepest node containing Q , the returned level is the minimum of the two bounds (cf. Figure 9(b) in the Appendix). **(3) Distribution.** By (2) and independence, $\Pr[\text{level}(Q) \geq \ell] = \Pr[\lambda_0(Q) \geq \lceil \ell/2 \rceil] \cdot \Pr[\lambda_1(Q) \geq \lfloor \ell/2 \rfloor]$ for $\ell \leq L^*$, and analogously with $\mu_i(Q)$; differences of consecutive tails give the stated distributions. For the c -2D-DAG, both factors equal 1 for $\ell \leq L^* - 2$, which yields the support $\{L^* - 2, L^* - 1, L^*\}$. $\square$

That π is a probability mass function follows from Lemma 3.4 below. The parameter η_i is the total length, around each point $mN_i/2^{\kappa_i}$ with $1 \leq m \leq 2^{\kappa_i} - 1$, of the starting positions x_i for which $Q^{(i)}$ is not contained in any depth- κ_i canonical interval; it vanishes when $s_i \leq \frac{c-2}{c-1} N_i/2^{\kappa_i}$, and we set $\eta_i = 0$ when the stated condition fails, since the corresponding event then holds with probability 1. For $\eta_0 = \eta_1 = 0$ the distribution is the reflection

Truncated quasi-dyadic distribution $\text{TQD}_c(\mathbf{N}, \mathbf{s})$

Parameters. Let $\mathbf{N} = (N_0, N_1)$ with $N_i = 2^{n_i+1}$, let $\mathbf{s} = (s_0, s_1)$ satisfy the shape assumptions, let $c = 2^\alpha + 1$ for an integer $\alpha \geq 1$, and let κ_i and L^* be as defined above. Put

$$\eta_0 = \max\{0, (c-1)s_0 - (c-2)N_0/2^{\kappa_0}\} \text{ if } 1 \leq \kappa_0 \leq \kappa_1 + 1,$$

$$\eta_1 = \max\{0, (c-1)s_1 - (c-2)N_1/2^{\kappa_1}\} \text{ if } 1 \leq \kappa_1 \leq \kappa_0,$$

and $\eta_i = 0$ otherwise.

Definition. $\text{TQD}_c(\mathbf{N}, \mathbf{s})$ is the distribution on $\{0, 1, \dots, L^*\}$ with probability mass function $\pi_k = W_k / ((N_0 - s_0)(N_1 - s_1))$, where $W_0 = (N_0 - s_0)(N_1 - s_1)$ if $L^* = 0$, and otherwise W_k is given as follows.

Case 1: $\kappa_0 \leq \kappa_1$ (implying $L^* = 2\kappa_0$):

$$k = 0 :$$

$$(N_0 - 2^{\kappa_0} s_0)(N_1 - 2^{\kappa_0} s_1) + 2^{\kappa_0-1} [(N_0 - 2^{\kappa_0} s_0) \eta_1 + (N_1 - 2^{\kappa_0-1} s_1) \eta_0];$$

$$k = 2j, 1 \leq j \leq \kappa_0 - 1 :$$

$$2^{\kappa_0-j} (s_0 - \eta_0)(N_1 - 2^{\kappa_0-j} s_1 - (2^{\kappa_0} - 2^{\kappa_0-j}) \eta_1) + 2^{\kappa_0-j-1} [(N_0 - 2^{\kappa_0-j} s_0 - (2^{\kappa_0} - 2^{\kappa_0-j}) \eta_0) \eta_1 + (N_1 - 2^{\kappa_0-j-1} s_1) \eta_0];$$

$$k = 2j + 1, 0 \leq j \leq \kappa_0 - 2 :$$

$$2^{\kappa_0-j-1} [(N_0 - 2^{\kappa_0-j} s_0 - (2^{\kappa_0} - 2^{\kappa_0-j}) \eta_0)(s_1 - \eta_1) + (2^{\kappa_0} - 2^{\kappa_0-j-1})(s_0 - \eta_0) \eta_1] + 2^{\kappa_0-j-2} (2^{\kappa_0} - 2^{\kappa_0-j-1}) s_1 \eta_0;$$

$$k = L^* - 1 :$$

$$(N_0 - 2s_0 - (2^{\kappa_0} - 2) \eta_0)(s_1 - \eta_1) + (2^{\kappa_0} - 1)(s_0 - \eta_0) \eta_1;$$

$$k = L^* :$$

$$(s_0 - \eta_0)(N_1 - s_1 - (2^{\kappa_0} - 1) \eta_1).$$

Case 2: $\kappa_0 > \kappa_1$ (implying $L^* = 2\kappa_1 + 1$):

$$k = 0 :$$

$$(N_0 - 2^{\kappa_1+1} s_0)(N_1 - 2^{\kappa_1} s_1) + 2^{\kappa_1} (N_1 - 2^{\kappa_1} s_1) \eta_0 + 2^{\kappa_1-1} (N_0 - 2^{\kappa_1} s_0) \eta_1;$$

$$k = 2j, 1 \leq j \leq \kappa_1 - 1 :$$

$$2^{\kappa_1-j} [(N_0 - 2^{\kappa_1-j+1} s_0 - (2^{\kappa_1+1} - 2^{\kappa_1-j+1}) \eta_0)(s_1 - \eta_1) + (N_1 - 2^{\kappa_1-j} s_1 - (2^{\kappa_1} - 2^{\kappa_1-j}) \eta_1) \eta_0] + 2^{\kappa_1-j-1} (N_0 - 2^{\kappa_1-j} s_0) \eta_1;$$

$$k = 2j + 1, 0 \leq j \leq \kappa_1 - 1 :$$

$$2^{\kappa_1-j} (s_0 - \eta_0)[N_1 - 2^{\kappa_1-j} s_1 - (2^{\kappa_1} - 2^{\kappa_1-j}) \eta_1] + 2^{\kappa_1-j-1} [(2^{\kappa_1+1} - 2^{\kappa_1-j}) (s_1 - \eta_1) \eta_0 + (2^{\kappa_1} - 2^{\kappa_1-j-1}) s_0 \eta_1];$$

$$k = L^* - 1 \text{ (if } \kappa_1 \geq 1) :$$

$$(N_0 - 2s_0 - (2^{\kappa_1+1} - 2) \eta_0)(s_1 - \eta_1) + (N_1 - s_1 - (2^{\kappa_1} - 1) \eta_1) \eta_0;$$

$$k = L^* :$$

$$(s_0 - \eta_0)(N_1 - s_1 - (2^{\kappa_1} - 1) \eta_1).$$

$k \mapsto L^* - k$ of the level distribution of the 2D-Tree (Lemma 3.2). Fig. 3 shows $\text{TQD}_3(\mathbf{N}, \mathbf{s})$ for a concrete choice of parameters.

LEMMA 3.4 (LEVEL OVERHEAD DISTRIBUTION). Assume uniform occupancy, dyadic domain and the shape assumptions, let $c = 2^\alpha + 1$ for an integer $\alpha \geq 1$, and let Q be drawn from $\mathcal{I}_s$. Then $\text{level}_{\text{diff}}(Q)$ has the distribution $\text{TQD}_c(\mathbf{N}, \mathbf{s})$ defined in the figure above; in particular, $\text{level}_{\text{diff}}(Q) \in \{0, \dots, L^*\}$.

PROOF SKETCH OF LEMMA 3.4. The proof has three steps.

(1) Joint law in each dimension. By Lemma 3.3, only the events $\{\mu_0(Q) \geq \lceil L^*/2 \rceil\}$ and $\{\mu_1(Q) \geq \lfloor L^*/2 \rfloor\}$ matter for the c -2D-DAG. Using the canonical windows from the proof of Lemma 3.3, the complement of each event consists of $c-1$ gaps just before each point $mN_i/2^{\kappa_i}$, $1 \leq m \leq 2^{\kappa_i} - 1$, of total length η_i per point (the event holds with probability 1 if $\eta_i = 0$). Such a point prevents containment of $Q^{(i)}$ in a depth- d dyadic interval only if it is a depth- d boundary. Together with (3), this gives, for $0 \leq d < \kappa_i$, that $\lambda_i(Q) = d$ outside the event with probability $2^d \eta_i / (N_i - s_i)$ and inside it with probability $2^d (s_i - \eta_i) / (N_i - s_i)$, while $\lambda_i(Q) = \kappa_i$ holds inside the event with probability $(N_i - 2^{\kappa_i} s_i) / (N_i - s_i)$.

(2) Decomposition by returned levels. The c -2D-DAG returns L^* , $L^* - 1$, or $L^* - 2$, according to which of the two events in (1) hold, and the 2D-Tree returns $2\lambda_0(Q)$ if $\lambda_0(Q) \leq \lambda_1(Q)$ and $2\lambda_1(Q) + 1$ otherwise (Lemmas 3.2 and 3.3). Hence, for every k , $\{\text{level}_{\text{diff}}(Q) = k\}$ is a disjoint union of at most three events, one per level returned by the c -2D-DAG, each the intersection of an event of dimension 0 and an event of dimension 1.

(3) Products. By independence of the two dimensions, each such event has probability equal to a product of two factors from (1), which gives the stated formulas. In particular, $\eta_0 = \eta_1 = 0$ if and only if the c -2D-DAG returns L^* with probability 1; then $\Pr[\text{level}_{\text{diff}}(Q) = k] = \Pr[\text{level}_{\text{Tree}}(Q) = L^* - k]$, as given by Lemma 3.2. $\square$

3.2 2D-Tree vs. c -2D-DAG: Stream of Queries

Section 3.1 gives the Level Overhead Distribution for a single random query of a fixed shape $\mathbf{s}$ (Lemma 3.4). We now use it to bound the two stream measures of Section 2.2, the average Returned Level Overhead $\overline{\text{level}_{\text{diff}}}$ and the average Accuracy Ratio $\overline{AR}$, first for a stream whose queries share a single shape (Theorem 3.5), and then for a stream that mixes several shapes (Corollary 3.6).

3.2.1 Uniform Query Shapes. Let $Q_1, \dots, Q_\tau$ be a stream of queries, each drawn from $\mathcal{I}_s$ for the same shape $\mathbf{s}$. By linearity of expectation, $\mathbb{E}[\overline{\text{level}_{\text{diff}}}] = \mathbb{E}_{\mathcal{I}_s}[\text{level}_{\text{diff}}(Q)]$ and $\mathbb{E}[\overline{AR}] = \mathbb{E}_{\mathcal{I}_s}[AR(Q)]$, whether or not the queries are independent. Hence it suffices to analyze a single query drawn from $\mathcal{I}_s$.

THEOREM 3.5 (STREAM PERFORMANCE, UNIFORM SHAPES). Assume uniform occupancy, dyadic domain, and the shape assumptions, let $c = 2^\alpha + 1$ for an integer $\alpha \geq 1$, and let $Q_1, \dots, Q_\tau$ be a stream of queries, each drawn from $\mathcal{I}_s$. Then:

(i) Average Returned Level Overhead.

$$\mathbb{E}[\overline{\text{level}_{\text{diff}}}] < \frac{(c-2)(5c-3)}{(c-1)^2},$$

independently of $N_0, N_1, \mathbf{s}$, and τ .

(ii) Average Accuracy Ratio.

$$\mathbb{E}[\overline{AR}] \geq \max\left\{1, 2^{\min\{\kappa_0, \kappa_1\}-1} - \frac{\min\{\kappa_0, \kappa_1\} + 2}{4}\right\} > \frac{1}{4} \min\left\{\frac{N_0}{s_0}, \frac{N_1}{s_1}\right\} - \frac{\min\{\kappa_0, \kappa_1\} + 2}{4}.$$

PROOF SKETCH OF THEOREM 3.5. It suffices to consider a single query Q drawn from $\mathcal{I}_s$.

(i) Since both returned levels lie in $\{0, \dots, L^*\}$, Lemma 3.3 gives $\mathbb{E}[\text{level}_{\text{diff}}(Q)] = \sum_{\ell=1}^{L^*} (1 - \Pr[\text{level}_{\text{Tree}}(Q) \geq \ell]) - (1 - p_*) - (1 - p_0 p_1)$, where $p_* = p_0$ if L^* is even and $p_* = p_1$ otherwise. Let $\kappa_0 \leq \kappa_1$ (the case $\kappa_0 > \kappa_1$ is similar) and $\rho_i = 2^{\kappa_0} s_i / N_i$. By Lemma 3.2 and (3), the 2D-Tree deficits sum to at most $(1 - 2^{-\kappa_0})(4\rho_0 + 3\rho_1 - 2\rho_0 \rho_1)$, and $1 - p_i \geq (1 - 2^{-\kappa_0})(c-1)(\rho_i - \frac{c-2}{c-1})^+$

by Lemma 3.3. The resulting function of (ρ_0, ρ_1) is maximized at $\rho_0 = \rho_1 = \frac{c-2}{c-1}$, where it equals $\frac{(c-2)(5c-3)}{(c-1)^2}$.

(ii) Let $\kappa_0 \leq \kappa_1$; the other case is symmetric. For $0 \leq a < \kappa_0$, on the event $\lambda_0(Q) = a \leq \lambda_1(Q)$, the 2D-Tree returns level $2a$, whereas the c -2D-DAG returns at least $2\kappa_0 - 1$ if $\mu_0(Q) = \kappa_0$ and $2\kappa_0 - 2$ otherwise. Weighting these events by their probabilities (Step 1 of the proof of Lemma 3.4) and summing the resulting geometric series over a gives the bound; the last inequality follows from $N_i/2^{\kappa_i+1} < s_i$. $\square$

The bound in (i) is tight: for $\kappa_0 = \kappa_1 = \kappa$ and $s_i \downarrow \frac{c-2}{c-1} \cdot N_i/2^\kappa$ (from above, so that $\lfloor \log_2(N_i/s_i) \rfloor = \kappa$), the expectation tends to $\frac{(c-2)(5c-3)}{(c-1)^2}$ as $\kappa \rightarrow \infty$. It increases from 3 for $c = 3$ to 5 as $c \rightarrow \infty$, and is the two-dimensional counterpart of the bound $2 \cdot \frac{c-2}{c-1}$ of the one-dimensional analysis [21]. Theorem 3.5 thus reveals an asymmetry between the two measures (Figure 4). Observe that both curves decrease with the side length s only on the whole: $\mathbb{E}[\text{level}_{\text{diff}}]$ follows a sawtooth pattern with one tooth per block $(N_i/2^{\kappa+1}, N_i/2^\kappa]$, never exceeds 2.89, 3.97 and 4.42 for $c = 3, 5, 9$ (the bounds are 3, 4.125 and 4.59), and vanishes at $s = N_i/2$, where both structures almost surely return the root, while $\mathbb{E}[\overline{AR}]$ decreases roughly like $1/s$ and is larger for larger c . Thus the expected Returned Level Overhead is bounded by a constant, whereas the expected Accuracy Ratio grows at least linearly in $\min\{N_0/s_0, N_1/s_1\}$, rather than logarithmically as in one dimension. The reason is that the 2D-Tree splits its root at the midpoint of dimension 0: whenever $Q^{(0)}$ contains this midpoint in its interior, which has probability $s_0/(N_0 - s_0)$, the 2D-Tree returns its root, whereas the c -2D-DAG never returns a level below $L^* - 2$. Such events contribute little to $\overline{\text{level}_{\text{diff}}}$, but each of them contributes a factor of at least 2^{L^*-2} to $\overline{AR}$. The same happens one dimension-depth deeper with a geometrically decreasing probability, and summing these contributions yields the linear growth.

3.2.2 Mixed Query Shapes. In practice, a query stream need not use a single shape. In the mixed-shape model of Section 2, the stream uses $\sigma \geq 1$ distinct shapes $\mathbf{s}_j = (s_j^{(0)}, s_j^{(1)})$ with frequencies $\omega_1, \dots, \omega_\sigma > 0$, $\sum_{j=1}^\sigma \omega_j = 1$: each query independently selects shape $\mathbf{s}_j$ with probability ω_j and then draws its starting position from $\mathcal{I}_{\mathbf{s}_j}$. For $j \in \{1, \dots, \sigma\}$, let $\kappa_j^{(i)} = \lfloor \log_2(N_i/s_j^{(i)}) \rfloor$ for $i \in \{0, 1\}$, $m_j = \min\{\kappa_j^{(0)}, \kappa_j^{(1)}\}$, and $\bar{m} = \sum_{j=1}^\sigma \omega_j m_j$.

COROLLARY 3.6 (STREAM PERFORMANCE, MIXED SHAPES). *Assume uniform occupancy and dyadic domain, let $c = 2^\alpha + 1$ for an integer $\alpha \geq 1$, and let $Q_1, \dots, Q_\tau$ be a stream of queries in the mixed-shape model, where every shape $\mathbf{s}_j$ satisfies the shape assumptions. Then:*

(i) Average Returned Level Overhead.

$$\mathbb{E}[\overline{\text{level}_{\text{diff}}}] < \frac{(c-2)(5c-3)}{(c-1)^2}.$$

The bound also holds if each Q_t is drawn from $\mathcal{I}_{\mathbf{s}(t)}$ for an arbitrary fixed sequence of shapes $\mathbf{s}(1), \dots, \mathbf{s}(\tau)$ satisfying the shape assumptions.

(ii) Average Accuracy Ratio.

$$\begin{aligned} \mathbb{E}[\overline{AR}] &\geq \sum_{j=1}^\sigma \omega_j \max\left\{1, 2^{m_j-1} - \frac{m_j+2}{4}\right\} \\ &\geq \max\left\{1, 2^{\bar{m}-1} - \frac{\bar{m}+2}{4}\right\}. \end{aligned}$$

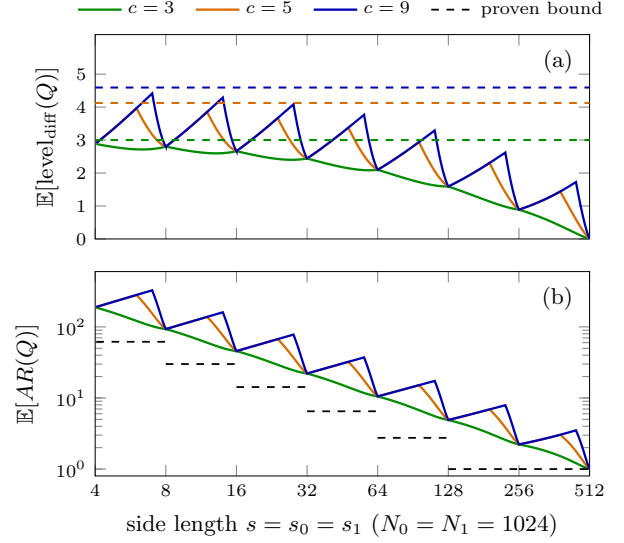


Figure 4: Illustration of Theorem 3.5 for $N_0 = N_1 = 1024$, square queries $s_0 = s_1 = s \in (4, 512]$, and $c \in \{3, 5, 9\}$. (a) The exact average Returned Level Overhead $\mathbb{E}[\text{level}_{\text{diff}}]$ and, dashed in the same colour, the bound $\frac{(c-2)(5c-3)}{(c-1)^2}$ of part (i), equal to 3, 4.125, and 4.59. (b) The exact average Accuracy Ratio $\mathbb{E}[\overline{AR}]$ (log scale) and the lower bound $\max\{1, 2^{\min\{\kappa_0, \kappa_1\}-1} - (\min\{\kappa_0, \kappa_1\} + 2)/4\}$ of part (ii), which changes at the powers of 2, where $\kappa_0 = \kappa_1$ changes. Neither quantity depends on the stream length τ .

PROOF SKETCH OF COROLLARY 3.6. By linearity of expectation and conditioning on the shape of each query, both averages are ω -weighted averages of single-shape expectations, each bounded by Theorem 3.5. Since the bound of Theorem 3.5(i) is the same for every shape, this gives (i), also for an arbitrary fixed sequence of shapes. Applying Theorem 3.5(ii) to each shape gives the first inequality in (ii). The second inequality follows from $\sum_j \omega_j \max\{1, g(m_j)\} \geq \max\{1, \sum_j \omega_j g(m_j)\}$ and Jensen's inequality for the convex function $g(x) = 2^{x-1} - (x+2)/4$. $\square$

The full proof and additional discussion are deferred to App. G.

4 Extension to Arbitrary Datasets via Distribution Learning

The analysis of Section 3 relies on uniform occupancy and a dyadic domain. We now present a general methodology that bridges this analysis to an arbitrary dataset $D \subseteq [M_0] \times [M_1]$, in which the $N \leq M_0 M_1$ elements of D may be heterogeneously distributed over the domain. The 2D-Tree, the c -2D-DAG, SRC-search, and both performance measures are defined in Sections 2.1 and 2.2 for an arbitrary dataset, so all of them remain well defined; what changes is the geometry of the canonical ranges.

What is preserved and what changes. Both structures are data-dependent (Section 2.1): whatever the distribution of D , every non-leaf node is split into children spanning $\lceil r/2 \rceil$ or $\lfloor r/2 \rfloor$ of its r distinct coordinates along the split dimension, so the number of splits along each dimension on any root-to-leaf path is at most $\lceil \log_2 N_i \rceil$, as under uniform occupancy. What changes is that sibling canonical ranges, while spanning (almost) the same number of coordinates, may have very different lengths: short where the elements are dense and long where they are sparse.

Two consequences follow. First, a random query of a fixed shape may descend along different paths than under uniform occupancy, so the returned level distributions of Lemmas 3.2 and 3.3 no longer apply and are in general analytically intractable. Second, unlike in one dimension, a child spanning half of the distinct coordinates of D_v along $i(v)$ need not hold half of the elements of D_v , because several elements may share a coordinate. Hence $|D_v|$ is no longer determined by $\text{level}(v)$, and the identity $AR(Q) = 2^{\text{level}_{\text{diff}}(Q)}$ of Section 3 holds only approximately. We quantify the deviation by the level-balance factor $\beta \geq 1$ of D , the smallest β such that every node v of the c -2D-DAG, and hence of the embedded 2D-Tree, satisfies

$$\frac{N}{\beta 2^{\text{level}(v)}} \leq |D_v| \leq \frac{\beta N}{2^{\text{level}(v)}}.$$

By Lemma 3.1(ii), $\beta = 1$ under uniform occupancy and a dyadic domain. In general, applying the lower bound to the node returned by the 2D-Tree and the upper bound to the node returned by the c -2D-DAG, and recalling $AR(Q) \geq 1$ (Section 2.2), gives for every query Q

$$AR(Q) \geq \max\{1, 2^{\text{level}_{\text{diff}}(Q)}/\beta^2\}. \quad (5)$$

Uniform baseline LODs. Since the individual returned level distributions are intractable, we focus on the Level Overhead Distribution (LOD) of D for a shape s , i.e., the distribution of $\text{level}_{\text{diff}}(Q)$ for Q drawn from $\mathcal{I}_s$, which combines both structures and, as Section 5 shows on real data, is well approximated by the uniform model even when the individual distributions are not. The approximating family is given by Lemma 3.4: for a dyadic $N^* = (N_0^*, N_1^*)$ and a shape s^* satisfying the shape assumptions for N^* , the distribution $\text{TQD}_c(N^*, s^*)$, with probability mass function π_k^* , is the LOD of a uniformly occupied dyadic domain of size N^* for queries of shape s^* . We call it a baseline LOD. The natural choice of N^* is the smallest dyadic size that accommodates D , $N_i^* = 2^{\lceil \log_2 N_i \rceil}$, and the natural choice of s^* is the nominal shape $s_i^* = s_i N_i^* / M_i$, which rescales dimension i from M_i to N_i^* positions; however, the best approximation need not be attained at the nominal shape.

Learning the closest baseline. Given D and s , we estimate the LOD of D by drawing sample queries from $\mathcal{I}_s$ and running SRC-search on both structures, and we select the baseline π^* that minimizes the L_2 distance to the empirical LOD over a set of candidate shapes s^* around the nominal shape. By the triangle inequality, the minimal distance estimates the distance ε between the LOD of D and π^* in the corollary below, up to a sampling error that vanishes as the number of sampled queries grows. The corollary transfers the guarantees of Theorem 3.5 to D .

COROLLARY 4.1 (TRANSFER TO ARBITRARY DATASETS). *Let D be an arbitrary dataset with level-balance factor β , let $c = 2^\alpha + 1$ for an integer $\alpha \geq 1$, and let $Q_1, \dots, Q_\tau$ be a stream of queries, each drawn from $\mathcal{I}_s$. Let $\pi^* = \text{TQD}_c(N^*, s^*)$ be a baseline LOD, let $m^* = \min_{i \in \{0,1\}} \lfloor \log_2(N_i^*/s_i^*) \rfloor$, and assume that*

$$\sum_k \left(\Pr_{\mathcal{I}_s}[\text{level}_{\text{diff}}(Q) = k] - \pi_k^* \right)^2 \leq \varepsilon^2,$$

where here and below the sums range over the integers k in the union of the supports of the two distributions. Then:

- (i) $\mathbb{E}[\text{level}_{\text{diff}}] < \frac{(c-2)(5c-3)}{(c-1)^2} + \varepsilon \left(\sum_k k^2 \right)^{1/2}$;
- (ii) $\mathbb{E}[AR] \geq \max\left\{1, \beta^{-2} \left(\max\left\{1, 2^{m^*-1} - \frac{m^*+2}{4}\right\} - \varepsilon \left(\sum_k 4^k \right)^{1/2} \right)\right\}$.

PROOF. By linearity of expectation, it suffices to consider a single query Q drawn from $\mathcal{I}_s$. For every function f on the integers, the Cauchy-Schwarz inequality and the assumption give

$$\left| \mathbb{E}[f(\text{level}_{\text{diff}}(Q))] - \sum_k f(k) \pi_k^* \right| \leq \varepsilon \left(\sum_k f(k)^2 \right)^{1/2}.$$

By Lemma 3.4, π^* is the LOD of the uniform model with parameters N^* and s^* , in which $AR = 2^{\text{level}_{\text{diff}}}$. Hence Theorem 3.5, applied with $\tau = 1$, gives $\sum_k k \pi_k^* < \frac{(c-2)(5c-3)}{(c-1)^2}$ and $\sum_k 2^k \pi_k^* \geq \max\{1, 2^{m^*-1} - (m^*+2)/4\}$. Taking $f(k) = k$ proves (i). Taking $f(k) = 2^k$ and using (5), which gives $\mathbb{E}[AR(Q)] \geq \mathbb{E}[2^{\text{level}_{\text{diff}}(Q)}]/\beta^2$ and $\mathbb{E}[AR(Q)] \geq 1$, proves (ii). $\square$

For a uniformly occupied dyadic domain, taking $N^* = (N_0, N_1)$ and $s^* = s$ gives $\varepsilon = 0$ and $\beta = 1$, and Corollary 4.1 reduces to Theorem 3.5. If both supports lie in $\{0, \dots, K\}$, the two sums are at most $K(K+1)(2K+1)/6$ and $(4^{K+1} - 1)/3$. Both error terms grow with the largest level overhead in the supports, and part (ii) also degrades with β ; the corollary is therefore informative when the empirical LOD is close to a baseline with a short support and the data are reasonably balanced. Section 5 reports, for the Gowalla dataset, the fitted baselines and their distances together with the two measures themselves.

5 Experimental Evaluation

We evaluate the 2D-Tree and the 3-2D-DAG on real two-dimensional data and ask how closely the uniform analysis of Section 3 describes them when its assumptions do not hold (Sec. 4). We present two shapes: $(5^\circ, 5^\circ)$ and $(5^\circ, 10^\circ)$; the summary table for all four shapes, additional details and plots are deferred to Appendix H.

5.1 Setup

Dataset. Gowalla is a location-based social network whose public check-in data [10] were collected between February 2009 and October 2010. The check-in file contains 6,442,892 records (user, UTC time, latitude, longitude, location id) from 107,092 users at 1,280,969 distinct locations. We discard 168 records with out-of-range coordinates or the placeholder position $(0^\circ, 0^\circ)$, leaving 1,256,676 distinct (latitude, longitude) pairs, and take as element set D the first $N = 2^{20} = 1,048,576$ of them, ordered by the time of their earliest check-in (4 Feb. 2009 to 8 Sep. 2010). We use the continuous embedding of Section 2, with latitude as dimension 0 on $[-90^\circ, 90^\circ)$ and longitude as dimension 1 on $[-180^\circ, 180^\circ)$; since both structures depend only on the ranks of the coordinates, no discretization of the domain is needed. D realizes $N_0 = 1,040,758$ distinct latitudes and $N_1 = 1,040,515$ distinct longitudes. The data are strongly non-uniform: the median latitude is 40.344° , against 0° for uniformly spread points.

Query streams. For each shape $s = (s_0, s_1) \in \{(5^\circ, 5^\circ), (5^\circ, 10^\circ), (10^\circ, 10^\circ), (25^\circ, 20^\circ)\}$ (latitude $\times$ longitude), the stream consists of queries $Q = [x_0, x_0 + s_0] \times [x_1, x_1 + s_1]$ whose starting position $x = (x_0, x_1)$ is chosen uniformly at random among all valid starting positions, i.e., the query stream model of Section 2 with a fixed shape. We keep only non-empty queries: we draw queries i.i.d., compute $|D \cap Q|$ exactly, and discard the empty ones (a fraction 0.815, 0.751, 0.665, and 0.352 of the draws, respectively). Accuracy, and thus the Accuracy Ratio as a false-positive factor, is meaningful only when $D \cap Q \neq \emptyset$ (Section 2.2). We issue an initial batch of 5,000 queries and then batches of 500, and compare the empirical Level Overhead Distributions (LODs) of the last two snapshots. We declare the stream stable when their L_2 distance drops below $\gamma = 0.001$ (Fig. 8), and then issue 120,000

further queries. All measures are computed on the final snapshot; the boxes in Fig. 7 cover all snapshots from stabilization on. The random seed is fixed, and per-query records are saved.

Uniform baseline and measures. For each shape we compare the empirical LOD with the distribution $\text{TQD}_3(N^*, s^*)$ of Lemma 3.4 for a uniformly occupied dyadic domain of the same size, $N^* = (2^{20}, 2^{20})$, i.e., $n_0 = n_1 = 19$. The nominal shape converts degrees into positions of this domain, $s_0^* = \text{round}(2^{20}s_0/180^\circ)$ and $s_1^* = \text{round}(2^{20}s_1/360^\circ)$, e.g., $(29, 127, 14, 564)$ for $(5^\circ, 5^\circ)$. We use the pair s^* within ± 500 of the nominal values (step 10) that minimizes the L_2 distance to the empirical LOD; every fitted s^* satisfies the shape assumptions of Lemma 3.4, and L^* below refers to it. We report $\text{level}_{\text{diff}}$ and $\overline{AR}$ over the non-empty queries, together with the median and the geometric mean of $\overline{AR}$. The reference values are the expectations of the fitted baseline, the bound of Theorem 3.5(i), which equals 3 for $c = 3$, and the lower bound of Theorem 3.5(ii) at the fitted s^* .

5.2 Results

Returned levels. Fig. 5 shows the empirical distributions of the returned levels with their cumulative distribution functions. The 3-2D-DAG returns deeper nodes than the 2D-Tree for every shape: the mean level is 9.45 against 6.75 for $(5^\circ, 5^\circ)$ and 9.10 against 6.50 for $(5^\circ, 10^\circ)$, and the CDF of the 3-2D-DAG never lies above that of the 2D-Tree. Smaller queries are answered at deeper levels. Because the data are non-uniform, we do not compare these distributions with the baseline.

Level Overhead Distribution. Figure 6 compares the empirical LOD with the best-fit baseline. In all 537,000 queries the 3-2D-DAG returns a node at least as deep as the 2D-Tree, i.e., $k = \text{level}_{\text{diff}} \geq 0$, although Section 2.1 guarantees this only under uniform occupancy and a dyadic domain. For the two shapes shown the baseline reproduces the shape of the LOD, including the larger mass at even than at odd k , with L_2 distances 0.057 and 0.085 (nominal parameters: 0.065, 0.090). For $(25^\circ, 20^\circ)$ the fit is poor ($L_2 = 0.316$); Appendix H discusses it.

Stream measures. Figure 7 gives the two stream measures, which Table 3 (Appendix H) lists for all four shapes. The average Returned Level Overhead is 2.69 and 2.60 for the shapes shown (0.87 to 2.69 over all four), below the bound 3 of Theorem 3.5(i) for every shape, within 0.53 of the baseline expectation, and stable (narrow boxes). It decreases as the queries grow, as does the baseline expectation in Fig. 4. Thus the overhead in traversal depth remains a small constant on real data.

The average Accuracy Ratio is 305.9 and 355.9 for the shapes shown (21.2 to 355.9 over all four). It exceeds both the baseline expectation and the lower bound of Theorem 3.5(ii). The typical query behaves differently: the median is 4 for both 5° shapes (1 to 4 over all four), and the geometric mean 6.46 and 6.07. In the data-dependent structures each level roughly halves $|D_v|$, so $\overline{AR}(Q)$ stays within 53% of $2^{\text{level}_{\text{diff}}(Q)}$ for every query, and the mean is dominated by rare large k : queries with $k > L^*$, which the baseline cannot produce, are 1.9% and 2.0% of the stream for the shapes shown, yet they carry 87% and 89% of the sum of $\overline{AR}$ values (Appendix H for the remaining shapes). These queries are small in rank space, so the 2D-Tree returns a shallow node because the query crosses an early split line, while the 3-2D-DAG descends much deeper; this is the root-return mechanism of Appendix H.2 acting at deeper levels, and it reflects that a fixed shape in degrees is a mixture of shapes in rank space (cf. Corollary 3.6).

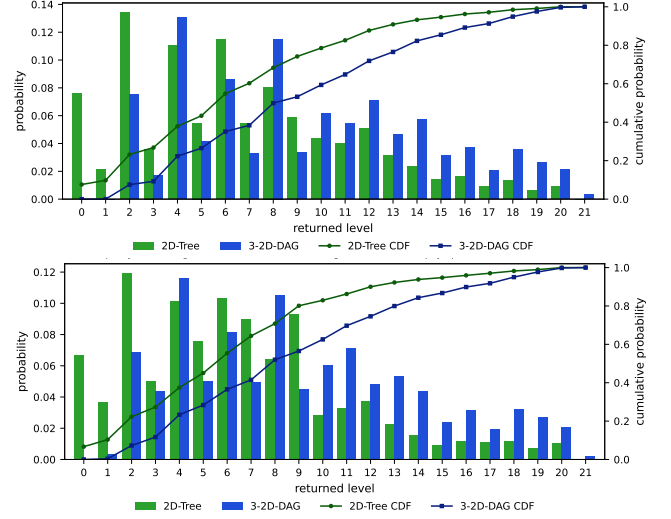


Figure 5: Returned level distributions on Gowalla, for the query shapes $(5^\circ, 5^\circ)$ and $(5^\circ, 10^\circ)$ (latitude $\times$ longitude). Green and blue bars: empirical probabilities for the 2D-Tree and the 3-2D-DAG (left axis); lines: their cumulative distribution functions (right axis). The 2D-Tree returns the root for 7.6% and 6.7% of the queries (Appendix H.2). The remaining two shapes appear in Fig. 12.

Stabilization. Figure 8 shows that the L_2 distance between consecutive LODs decreases with the number of queries and falls below γ after 18,500 and 16,000 non-empty queries for the two shapes shown (9,500 and 13,000 for the remaining shapes). The distance continues to decrease beyond this point, to about 10^{-4} at the end of the streams, so the additional 120,000 queries are well past stabilization.

In summary, on Gowalla the 3-2D-DAG keeps the level overhead below the proven constant, and it reduces the number of returned elements by a factor of about 2^k that is moderate for typical queries (4 at the median for the 5° shapes) and very large for the rare queries on which the 2D-Tree stops early.

6 Privacy Analysis

In SRC-based searchable encryption the server learns which canonical range, hence which element set D_v , answers a query [12, 15]. If the search is deterministic, repeated executions of the same query Q return the same node, so a persistent adversary can observe how often each token occurs and relate it to the structure of the DAG [15]; such frequency information is the input of leakage-abuse attacks [19, 22]. We show that the overlap of the c -2D-DAG can be used to blur this, at no cost in accuracy.

Random maximal rule. For a query Q , let $M(Q)$ be the set of nodes v with $Q \subseteq \text{range}_v$ that attain the minimum of $|D_v|$ in the definition of SRC-search (Section 2.1). The **random maximal rule** runs the exhaustive SRC-search, collects all of $M(Q)$, and returns one of its nodes uniformly at random. All nodes of $M(Q)$ have the same $|D_v|$, so the rule changes neither $|D_{v_{\mathcal{G}}(Q)}|$ nor $\overline{AR}(Q)$ (Section 2.2). It can change the returned level only if $M(Q)$ contains nodes on different levels; this cannot happen under the assumptions of Section 3, and it happens for only 0.4% of the Gowalla queries (the experiments of Section 5.1 break such ties in favor of the deepest node). For a fixed Q we measure the uncertainty about the returned node by its **Shannon entropy**

$$H(Q) = \log_2 |M(Q)| \quad (\text{bits}).$$

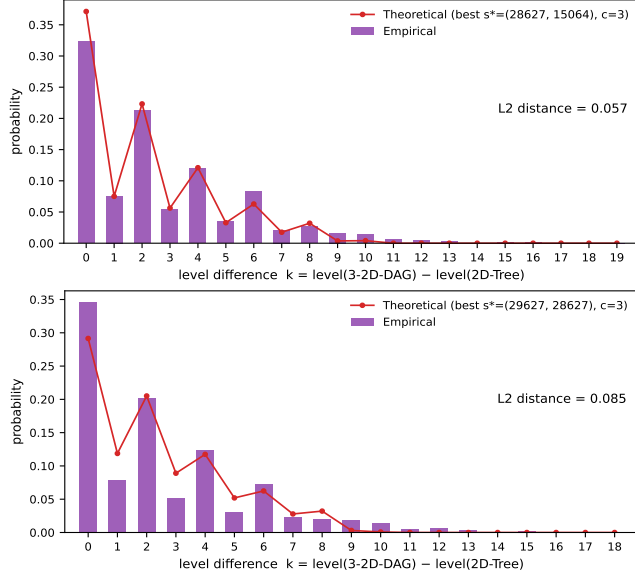


Figure 6: Level Overhead Distributions on Gowalla, for the query shapes $(5^\circ, 5^\circ)$ and $(5^\circ, 10^\circ)$. Purple bars: empirical LOD of $k = \text{level}_{\text{diff}}$ over non-empty queries. Red line: $\text{TQD}_3(N^*, s^*)$ of Lemma 3.4 at the best-fit s^* (given in each panel), with its L_2 distance to the empirical LOD. The remaining two shapes appear in Fig. 13.

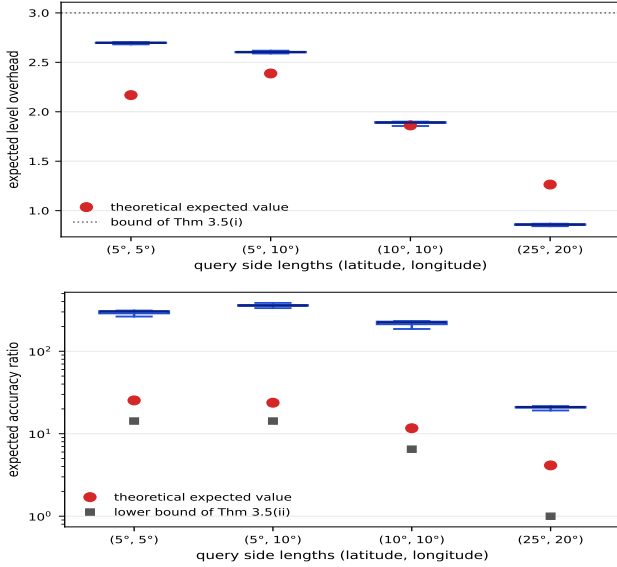


Figure 7: Average Returned Level Overhead (left) and average Accuracy Ratio (right, logarithmic axis) on Gowalla. Boxes: snapshots from stabilization on. Red dots: expectations of the best-fit baseline. Dotted line: bound of Theorem 3.5(i). Grey squares: lower bound of Theorem 3.5(ii).

A positive $H(Q)$ means that repeated executions of Q may disclose different canonical ranges. The following is proved in App. I.

- PROPOSITION 6.1.** (i) *On the 2D-Tree, $|M(Q)| = 1$, hence $H(Q) = 0$, for every dataset and every query Q .*
- (ii) *Assume uniform occupancy and a dyadic domain, and let $c = 2^\alpha + 1$. On the c -2D-DAG, $1 \leq |M(Q)| \leq (c - 1)^2$, hence $0 \leq H(Q) \leq 2\alpha$, for every query Q .*

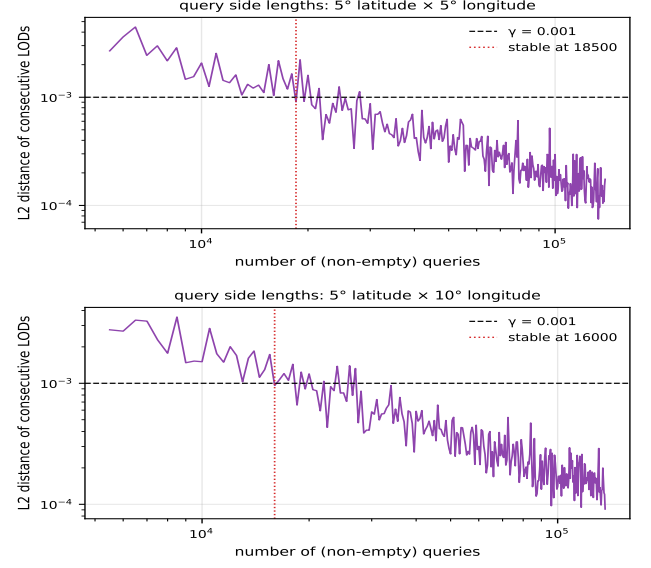


Figure 8: L_2 distance between the LODs of consecutive snapshots against the number of non-empty queries, for the query shapes $(5^\circ, 5^\circ)$ and $(5^\circ, 10^\circ)$. Dashed line: $\gamma = 0.001$; dotted line: stabilization point. The remaining two shapes appear in Fig. 14.

Table 2: Gowalla, 3-2D-DAG with the random maximal rule (2D-Tree: $H = 0$). Node level: exact over all non-empty queries. Element-set level: entropy of D_v (nodes with equal element sets merged), estimated on 1,200 queries with $|M(Q)| > 1$ per shape (95% confidence interval at most ± 0.008 bits).

Shape	$\Pr[M > 1]$	mean H	max $ M $	$\Pr[H_{\text{set}} > 0]$	mean H_{set}
(5, 5)	20.48%	0.351	168	16.5%	0.218
(5, 10)	17.25%	0.296	168	14.2%	0.185
(10, 10)	12.97%	0.213	162	10.2%	0.132
(25, 20)	6.42%	0.073	8	5.2%	0.056

- (iii) *The upper bound in (ii) is not trivial: for $c = 3$, the query Q_1 of Fig. 2 has $|M(Q_1)| = 2$, so $H(Q_1) = 1$.*

Gowalla. Table 2 reports $|M(Q)|$ on the non-empty query streams of Section 5 (sizes in Table 3). By Proposition 6.1(i), the 2D-Tree has $H(Q) = 0$ on every query. The c -2D-DAG built on Gowalla is not uniform, so the bound of Proposition 6.1(ii) does not apply: $|M(Q)|$ reaches 168. These large sets occur at deep levels where nodes hold few elements (for $|M(Q)| \geq 10$, the median $|D_v|$ is 4 at $(5^\circ, 5^\circ)$). Still, 20.5%, 17.3%, 13.0%, and 6.4% of the queries have $H(Q) > 0$, and the mean entropy is 0.35, 0.30, 0.21, and 0.07 bits, decreasing with the query size as the overhead and accuracy measures do (Section 5.2).

Remarks. Distinct nodes of $M(Q)$ may store the same element set, so the entropy of the returned set (last two columns of Table 2) is lower than the node entropy; the two coincide in the uniform model, where distinct rectangles hold distinct sets. The entropy is per execution: an adversary repeating Q eventually sees all of $M(Q)$, so the rule hinders but does not remove linkage of repeated queries. Finally, a query with $|M(Q)| = 1$ is as deterministic as on the 2D-Tree, so the gain concentrates on about one query in five at the smallest shape.

Artifacts

The source code and scripts used for the experiments, together with instructions for reproducing all results and figures, are available in a public repository at <https://github.com/ladan90/Data-Structures-Simulation/tree/main/2D>. The repository also contains the full version of this paper, including the appendix. The Gowalla dataset used in our experiments is publicly available from the Stanford Network Analysis Project (SNAP) at <https://snap.stanford.edu/data/loc-gowalla.html>.

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APPENDIX

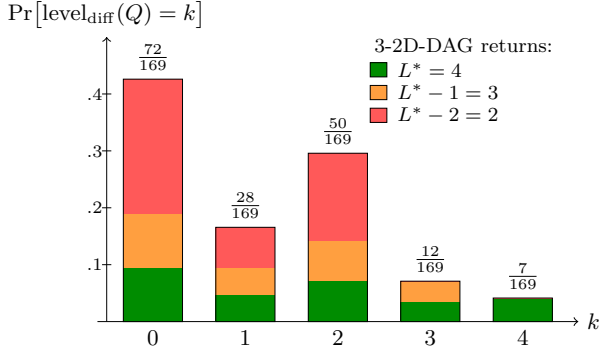


Figure 10: Illustration of Lemma 3.4 for $N_0 = N_1 = 16$, $s_0 = s_1 = 3$, and $c = 3$; thus $\kappa_0 = \kappa_1 = 2$, $L^* = 4$ (Case 1), and $\eta_0 = \eta_1 = 2$. Each bar gives $\Pr_{\mathcal{I}_k}[\text{level}_{\text{diff}}(Q) = k]$ and is split according to the level returned by the 3-2D-DAG, i.e., into the three disjoint events whose probabilities are summed in the formulas of Lemma 3.4. The 3-2D-DAG returns L^* , $L^* - 1$, and $L^* - 2$ with probabilities $49/169$, $42/169$, and $78/169$, respectively (Lemma 3.3). If $\eta_0 = \eta_1 = 0$, only the first event remains, and the LOD reduces to the reflected 2D-Tree distribution $\Pr[\text{level}_{\text{diff}}(Q) = k] = \Pr[\text{level}_{\text{Tree}}(Q) = L^* - k]$.

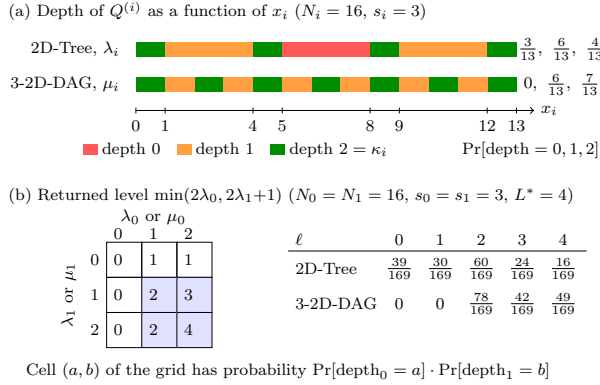


Figure 9: Illustration of Lemmas 3.2 and 3.3 for $N_0 = N_1 = 16$, $s_0 = s_1 = 3$, and $c = 3$. (a) Exact per-dimension containment depths as functions of the query start $x_i \in [0, 13]$. The shifted canonical intervals of the 3-2D-DAG remove the depth-0 outcome and increase the probability of the deepest depth. (b) The returned two-dimensional level is $\min(2d_0, 2d_1 + 1)$, where (d_0, d_1) is (λ_0, λ_1) for the 2D-Tree and (μ_0, μ_1) for the 3-2D-DAG. The right-hand table gives the resulting exact level distributions.

A Conclusions and Future Work

We introduced a competitive framework for comparing data structures supporting SRC-search with two-dimensional range queries, and used it to prove several advantages of the c -2D-DAG family over the 2D-Tree: an additive constant returned level overhead and a multiplicative gain in accuracy that grows with the inverse query size, at the price of a denser structure. Our main technical tool, the Level Overhead Distribution, stabilizes the

comparison where the individual returned level distributions become intractable, and a learned uniform baseline carries the guarantees over to arbitrary datasets, as confirmed empirically on the Gowalla check-in data. Randomizing the choice among equally accurate nodes further can yield positive entropy of the returned node, which the deterministic 2D-Tree lacks, at no cost in accuracy.

Several directions remain open. First, our methodology applies to other structures and search principles: comparing the c -2D-DAG against exact covering rules such as BRC and URC, against the range trees and quadrees accommodated by the framework of Falzon et al. [15], or against covers consisting of a constant number of nodes instead of one. Second, the behavior of the LOD under strongly biased data configurations, as observed for the largest query shape on Gowalla, calls for richer baselines, e.g., mixtures of uniform LODs, and for a deeper study of the level-balance factor. Third, extending the analysis to more than two search dimensions, to dynamic datasets and updates, and to stronger privacy measures, such as the entropy of the returned element set and other randomized selection rules, are natural next steps. Finally, it would be interesting to determine whether sparse structures exist that dominate the c -2D-DAG, and what trade-offs between search cost, accuracy, and leakage they admit.

B Proof of Lemma 3.1 from Section 3

PROOF. (i) By induction on the level. The root satisfies the claim. Let v be a non-leaf node satisfying it, with side $[a, a + L)$ along $i = i(v)$. Since D_v consists of all integer points of range v and the side of range v along $1 - i$ contains an integer, the distinct coordinates realized by D_v along i are $u_g = a + g - 1$ for $g = 1, \dots, r$, with $r = L$. A child spanning ranks g to h therefore has side $[a + g - 1, a + h)$ along i , an interval with integer endpoints, and its element set consists of all integer points of its canonical range. Since L is a power of 2 and $L = r \geq 2$, every child spans $\lceil r/2 \rceil = \lfloor r/2 \rfloor = L/2$ ranks, so its side along i has length $L/2$, again a power of 2. Under uniform occupancy, the number of elements of a canonical range with integer endpoints equals its area.

(ii) By (i), every child of a node has half the area of that node, so every node at level ℓ has area, and hence element count, $N_0 N_1 / 2^\ell$. A node is a leaf if and only if $|D_v| \leq 1$, i.e., if and only if $\ell = \log_2(N_0 N_1) = n_0 + n_1 + 2$.

(iii) Suppose that every level $\ell' < \ell$ has split dimension $\ell' \bmod 2$. Then a level ℓ node results from $\lceil \ell/2 \rceil$ splits along dimension 0 and $\lfloor \ell/2 \rfloor$ splits along dimension 1, each halving the corresponding side by (i), which gives the stated side lengths. Given these side lengths at level ℓ' , a node at level ℓ' has split dimension $\ell' \bmod 2$ exactly when its side along dimension $\ell' \bmod 2$ has length at least 2, i.e., when $\ell' \leq 2n_0$ for even ℓ' and $\ell' \leq 2n_1 + 1$ for odd ℓ' . All levels $\ell' < \ell$ satisfy this condition if and only if $\ell \leq \hat{\ell}$, which proves both claims by induction on ℓ .

(iv) By the proof of (i), $r = L$ and $u_g = a + g - 1$. Since L and $2(c - 1) = 2^{\alpha+1}$ are powers of 2, $L \geq 2(c - 1)$ implies $\lfloor r/(2(c - 1)) \rfloor = L/(2(c - 1))$. The augmented child w_{j+1} , $1 \leq j \leq c - 2$, spans ranks $jL/(2(c - 1)) + 1$ to $jL/(2(c - 1)) + L/2$, and w_1 and w_c span ranks 1 to $L/2$ and $L/2 + 1$ to L ; by the rank convention of Section 2.1, these are the stated intervals. If $L < 2(c - 1)$, then $\lfloor r/(2(c - 1)) \rfloor = 0$, and the augmented children coincide with w_1 (Section 2.1). $\square$

C Proof of Lemma 3.2 from Section 3.1

PROOF. Per-dimension depths. Fix $i \in \{0, 1\}$. Every depth- d dyadic interval is contained in a depth- $(d-1)$ dyadic interval, so containment of $Q^{(i)}$ at depth d implies containment at every depth $d' \leq d$, and $\{\lambda_i(Q) \geq d\}$ is the event that $Q^{(i)}$ is contained in some depth- d dyadic interval. Since $\lambda_i(Q)$ depends only on x_i , and x_0 and x_1 are independent under $\mathcal{I}_s$, $\lambda_0(Q)$ and $\lambda_1(Q)$ are independent. For $0 \leq d \leq \kappa_i$, the half-open interval $Q^{(i)} = [x_i, x_i + s_i]$ is contained in $[jN_i/2^d, (j+1)N_i/2^d)$ if and only if $x_i \in [jN_i/2^d, (j+1)N_i/2^d - s_i]$. These 2^d intervals of starting positions have length $N_i/2^d - s_i \geq 0$, are pairwise disjoint since $s_i > 0$, and lie in $[0, N_i - s_i]$. Since x_i is uniform on $[0, N_i - s_i]$, this proves (3).

Returned level. By Lemma 3.1(iii) and (iv) with $\alpha = 0$, and since $L^* + 1 \leq \hat{\ell}$, for every $\ell \leq L^* + 1$ the canonical ranges of the level ℓ nodes of the 2D-Tree are exactly the products $A \times B$ of a depth- $\lceil \ell/2 \rceil$ dyadic interval A of dimension 0 and a depth- $\lfloor \ell/2 \rfloor$ dyadic interval B of dimension 1: this holds at the root, and a level ℓ node with $\ell < \hat{\ell}$ is split along dimension $\ell \bmod 2$ at the midpoint of its side, which halves exactly one of the two factors. Hence, for $\ell \leq L^* + 1$, some level ℓ node contains Q if and only if $\lceil \ell/2 \rceil \leq \lambda_0(Q)$ and $\lfloor \ell/2 \rfloor \leq \lambda_1(Q)$, i.e., if and only if $\ell \leq 2\lambda_0(Q)$ and $\ell \leq 2\lambda_1(Q) + 1$. Since $\lambda_i(Q) \leq \kappa_i$, the largest such ℓ is $\min(2\lambda_0(Q), 2\lambda_1(Q) + 1) \leq L^*$, and no node at a level greater than L^* contains Q . Since SRC-search on the 2D-Tree returns the deepest node containing Q (Section 2.1), this proves (i) and the support claim of (ii).

Distribution. By (i), for $0 \leq \ell \leq L^*$, $\{\text{level}_{\text{Tree}}(Q) \geq \ell\} = \{\lambda_0(Q) \geq \lceil \ell/2 \rceil\} \cap \{\lambda_1(Q) \geq \lfloor \ell/2 \rfloor\}$, where $\lceil \ell/2 \rceil \leq \kappa_0$ and $\lfloor \ell/2 \rfloor \leq \kappa_1$ by the definition of L^* . By independence and (3),

$$\Pr[\text{level}_{\text{Tree}}(Q) \geq \ell] = \frac{(N_0 - 2^{\lceil \ell/2 \rceil} s_0)(N_1 - 2^{\lfloor \ell/2 \rfloor} s_1)}{(N_0 - s_0)(N_1 - s_1)}, \quad 0 \leq \ell \leq L^*.$$

For $\ell < L^*$, the claim follows from $\Pr[\text{level}_{\text{Tree}}(Q) = \ell] = \Pr[\text{level}_{\text{Tree}}(Q) \geq \ell] - \Pr[\text{level}_{\text{Tree}}(Q) \geq \ell + 1]$: for even ℓ , the two tails share the factor $N_1 - 2^{\ell/2} s_1$ and their factors along dimension 0 differ by $2^{\ell/2+1} s_0 - 2^{\ell/2} s_0 = 2^{\ell/2} s_0$; for odd ℓ , they share the factor $N_0 - 2^{(\ell+1)/2} s_0$ and their factors along dimension 1 differ by $2^{(\ell+1)/2} s_1 - 2^{(\ell-1)/2} s_1 = 2^{(\ell-1)/2} s_1$. For $\ell = L^*$, $\Pr[\text{level}_{\text{Tree}}(Q) = L^*] = \Pr[\text{level}_{\text{Tree}}(Q) \geq L^*]$. $\square$

D Proof of Lemma 3.3 from Section 3.1

PROOF. Per-dimension depths. Fix $i \in \{0, 1\}$. For $d \geq 1$, let $L = N_i/2^{d-1}$ and $\delta = L/(2(c-1))$, so that the depth- $(d-1)$ canonical intervals are $[2m\delta, 2m\delta + L)$, $m \in \{0, \dots, (c-1)(2^{d-1} - 1)\}$, and the depth- d canonical intervals are $[j\delta, j\delta + L/2)$, $j \in \{0, \dots, (c-1)(2^d - 1)\}$. The latter interval is contained in the former if and only if $2m \leq j \leq 2m + c - 1$, and, as m ranges over its index set, the indices $2m, \dots, 2m + c - 1$ cover $\{0, \dots, (c-1)(2^d - 1)\}$, since $c \geq 2$. Hence every depth- d canonical interval is contained in a depth- $(d-1)$ canonical interval, containment of $Q^{(i)}$ at depth d implies containment at every depth $d' \leq d$, and $\{\mu_i(Q) \geq d\}$ is the event that $Q^{(i)}$ is contained in some depth- d canonical interval. Since $\mu_i(Q)$ depends only on x_i , $\mu_0(Q)$ and $\mu_1(Q)$ are independent.

Fix $0 \leq d \leq \kappa_i$ and let $\delta_d = N_i/(2^d(c-1))$. The interval $Q^{(i)}$ is contained in the j -th depth- d canonical interval if and only if $x_i \in [j\delta_d, j\delta_d + N_i/2^d - s_i]$. These $(c-1)2^d - (c-2)$ windows have length $N_i/2^d - s_i \geq 0$, the first starts at 0, the last ends at $(c-1)(2^d - 1)\delta_d + N_i/2^d - s_i = N_i - s_i$, and consecutive windows are separated by a signed gap of $\delta_d - (N_i/2^d - s_i) = s_i - \frac{c-2}{c-1} \cdot \frac{N_i}{2^d}$. If

this gap is non-positive, the windows cover $[0, N_i - s_i]$; otherwise, they are pairwise disjoint. Since x_i is uniform on $[0, N_i - s_i]$, this proves (4). For $0 \leq d \leq \kappa_i - 1$, we have $s_i \leq N_i/2^{d+1} \leq \frac{1}{2} \cdot \frac{N_i}{2^d} \leq \frac{c-2}{c-1} \cdot \frac{N_i}{2^d}$, since $c \geq 3$; hence the windows cover $[0, N_i - s_i]$, and $\mu_i(Q) \geq \kappa_i - 1$ for every x_i .

Returned level. We show by induction on ℓ that, for every $\ell \leq L^*$, the canonical ranges of the level ℓ nodes of the c -2D-DAG are exactly the products $A \times B$ of a depth- $\lceil \ell/2 \rceil$ canonical interval A of dimension 0 and a depth- $\lfloor \ell/2 \rfloor$ canonical interval B of dimension 1. At $\ell = 0$, the only depth-0 canonical interval of dimension i is $[0, N_i]$. Let $\ell < L^*$. By Lemma 3.1(iii), every level ℓ node has split dimension $i = \ell \bmod 2$, and its factor along i is a depth- $(d-1)$ canonical interval $[2m\delta, 2m\delta + L)$ in the notation above, where $d-1 = \lceil \ell/2 \rceil$ if $i = 0$ and $d-1 = \lfloor \ell/2 \rfloor$ if $i = 1$. Since $\ell < L^*$, $L \geq 2(c-1)$ (Section 3), so by Lemma 3.1(iv) the children of the node have factors $[(2m+j')\delta, (2m+j')\delta + L/2)$, $j' = 0, \dots, c-1$, along i and keep the factor along $1-i$. The level $(\ell+1)$ nodes are the children of the level- ℓ nodes; by the induction hypothesis and the covering argument above, they are therefore exactly the products of a depth- d canonical interval along i with a canonical interval of unchanged depth along $1-i$. Since the depth along i increases by one and the depth along $1-i$ is unchanged, these depths are $\lceil (\ell+1)/2 \rceil$ along dimension 0 and $\lfloor (\ell+1)/2 \rfloor$ along dimension 1, which is the claim at level $\ell+1$. Hence, for $\ell \leq L^*$, some level- ℓ node contains Q if and only if $\lceil \ell/2 \rceil \leq \mu_0(Q)$ and $\lfloor \ell/2 \rfloor \leq \mu_1(Q)$, i.e., if and only if $\ell \leq 2\mu_0(Q)$ and $\ell \leq 2\mu_1(Q) + 1$. Since $\mu_i(Q) \leq \kappa_i$, the largest such ℓ is $\min(2\mu_0(Q), 2\mu_1(Q) + 1) \leq L^*$, and no node at a level greater than L^* contains Q (Section 3). Since SRC-search minimizes $|D_v|$ (Section 2.1) and $|D_v| = N/2^\ell$ for every node v at level ℓ (Lemma 3.1(ii)), SRC-search returns a deepest node containing Q , which proves (i).

Distribution. By (i), for $0 \leq \ell \leq L^*$, $\Pr[\text{level}_{c\text{-DAG}}(Q) \geq \ell] = \Pr[\mu_0(Q) \geq \lceil \ell/2 \rceil] \cdot \Pr[\mu_1(Q) \geq \lfloor \ell/2 \rfloor]$, and $\Pr[\mu_i(Q) \geq d] = 1$ for $0 \leq d \leq \kappa_i - 1$. For $\ell = L^*$, this tail is $p_0 p_1$. For $\ell = L^* - 1 \geq 0$, we have $(\lceil \ell/2 \rceil, \lfloor \ell/2 \rfloor) = (\lceil L^*/2 \rceil, \lfloor L^*/2 \rfloor - 1)$ if L^* is even and $(\lceil L^*/2 \rceil - 1, \lfloor L^*/2 \rfloor)$ if L^* is odd, so the tail is p_0 or p_1 , respectively. For $\ell = L^* - 2 \geq 0$, we have $(\lceil \ell/2 \rceil, \lfloor \ell/2 \rfloor) = (\lceil L^*/2 \rceil - 1, \lfloor L^*/2 \rfloor - 1)$, so the tail is 1. Taking differences of consecutive tails gives the stated probabilities and shows that $\text{level}_{c\text{-DAG}}(Q) \geq L^* - 2$. Finally, if $L^* = 0$, then $p_0 = \Pr[\mu_0(Q) \geq 0] = 1$ and $p_1 = \Pr[\mu_1(Q) \geq 0] = 1$; if $L^* = 1$, then $p_1 = \Pr[\mu_1(Q) \geq 0] = 1$. In both cases, the expressions for negative levels vanish. $\square$

E Proof of Lemma 3.4 from Section 3.1

PROOF. All probabilities are taken under $\mathcal{I}_s$. As in the proofs of Lemmas 3.2 and 3.3, the probability of an event that depends only on x_i is the total length of the corresponding starting positions divided by $N_i - s_i$. If $L^* = 0$, both structures return level 0 by Lemmas 3.2 and 3.3; assume $L^* \geq 1$. Since $\lambda_i(Q) \leq \mu_i(Q)$, we have $\text{level}_{\text{Tree}}(Q) \leq \text{level}_{c\text{-DAG}}(Q) \leq L^*$, hence $\text{level}_{\text{diff}}(Q) \in \{0, \dots, L^*\}$. Within this proof only, let $\theta_0 = \lceil L^*/2 \rceil$ and $\theta_1 = \lfloor L^*/2 \rfloor$; thus $(\theta_0, \theta_1) = (\kappa_0, \kappa_0)$ in Case 1 and $(\theta_0, \theta_1) = (\kappa_1 + 1, \kappa_1)$ in Case 2. In both cases, $\theta_i \leq \kappa_i$, and the condition in the definition of η_i is equivalent to $1 \leq \theta_i = \kappa_i$.

Step 1: gaps of the canonical windows. Fix $i \in \{0, 1\}$ and $0 \leq d \leq \kappa_i$. We show

$$\begin{aligned} \Pr[\lambda_i(Q) \geq d, \mu_i(Q) < \kappa_i] &= \\ &= \frac{(2^{\kappa_i} - 2^d) \max\{0, (c-1)s_i - (c-2)N_i/2^{\kappa_i}\}}{N_i - s_i}. \end{aligned}$$

Let $\delta = N_i/(2^{\kappa_i}(c-1))$. By the proof of Lemma 3.3, $Q^{(i)}$ is contained in the j -th depth- κ_i canonical interval if and only if $x_i \in [j\delta, j\delta + N_i/2^{\kappa_i} - s_i]$, for $j \in \{0, \dots, (c-1)(2^{\kappa_i}-1)\}$; these windows start at 0, end at $N_i - s_i$, and consecutive windows are separated by a signed gap of $s_i - (c-2)\delta$. If $s_i \leq (c-2)\delta$, the windows cover $[0, N_i - s_i]$, so $\mu_i(Q) = \kappa_i$ with probability 1 and both sides of (6) vanish. Otherwise, $\{\mu_i(Q) < \kappa_i\}$ is the union of the $(c-1)(2^{\kappa_i}-1)$ open gaps $(j\delta + N_i/2^{\kappa_i} - s_i, (j+1)\delta)$, each of length $s_i - (c-2)\delta$. For $m \in \{1, \dots, 2^{\kappa_i}-1\}$, the $c-1$ gaps with $(m-1)(c-1) \leq j \leq m(c-1)-1$ lie in $[mN_i/2^{\kappa_i} - s_i, mN_i/2^{\kappa_i}]$ and have total length $(c-1)s_i - (c-2)N_i/2^{\kappa_i}$. The open interval $(mN_i/2^{\kappa_i} - s_i, mN_i/2^{\kappa_i})$ is the set of starting positions for which $mN_i/2^{\kappa_i}$ lies in the interior of $Q^{(i)}$; these intervals are pairwise disjoint, since $s_i \leq N_i/2^{\kappa_i}$. Moreover, $Q^{(i)}$ is contained in a depth- d dyadic interval if and only if no point $mN_i/2^{\kappa_i}$ with m a multiple of 2^{κ_i-d} lies in the interior of $Q^{(i)}$. Hence, up to finitely many starting positions, $\{\lambda_i(Q) \geq d\} \cap \{\mu_i(Q) < \kappa_i\}$ is the union of the gaps associated with the $2^{\kappa_i} - 2^d$ values of m that are not multiples of 2^{κ_i-d} , which proves (6).

Step 2: joint law at the thresholds. For $i \in \{0, 1\}$ and $0 \leq d \leq \theta_i$,

$$\Pr[\lambda_i(Q) \geq d, \mu_i(Q) < \theta_i] = \frac{(2^{\theta_i} - 2^d) \eta_i}{N_i - s_i}. \quad (6)$$

If $1 \leq \theta_i = \kappa_i$, this is (6). Otherwise, $\eta_i = 0$ and $\mu_i(Q) \geq \theta_i$ with probability 1: if $\theta_i < \kappa_i$, because $\mu_i(Q) \geq \kappa_i - 1$ (Lemma 3.3); if $\theta_i = \kappa_i$, then $\mu_i(Q) \geq \theta_i - 1$, and for $d = 0$, (6) gives $1 - p_i = (2^{\theta_i} - 1)\eta_i/(N_i - s_i)$ for p_i as in Lemma 3.3(ii). Since $\Pr[\lambda_i(Q) = d] = 2^d s_i/(N_i - s_i)$ for $d < \kappa_i$ by (3), (6) yields, for $0 \leq d \leq \theta_i - 1$,

$$\begin{aligned} \Pr[\lambda_i(Q) = d, \mu_i(Q) < \theta_i] &= \frac{2^d \eta_i}{N_i - s_i}, \\ \Pr[\lambda_i(Q) = d, \mu_i(Q) \geq \theta_i] &= \frac{2^d (s_i - \eta_i)}{N_i - s_i}, \end{aligned}$$

and, for $0 \leq d \leq \theta_i$, by (3),

$$\Pr[\lambda_i(Q) \geq d, \mu_i(Q) \geq \theta_i] = \frac{N_i - 2^d s_i - (2^{\theta_i} - 2^d) \eta_i}{N_i - s_i}.$$

Step 3: returned levels. By Lemma 3.2(i), $\text{level}_{\text{Tree}}(Q) = 2\lambda_0(Q)$ if $\lambda_0(Q) \leq \lambda_1(Q)$, and $\text{level}_{\text{Tree}}(Q) = 2\lambda_1(Q) + 1$ otherwise. By Lemma 3.3(i) and $\mu_i(Q) \geq \theta_i - 1$, the c -2D-DAG returns L^* on $\{\mu_0(Q) \geq \theta_0, \mu_1(Q) \geq \theta_1\}$. In Case 1, it returns $L^* - 1$ on $\{\mu_0(Q) \geq \theta_0, \mu_1(Q) < \theta_1\}$ and $L^* - 2$ on $\{\mu_0(Q) < \theta_0\}$. In Case 2, it returns $L^* - 1$ on $\{\mu_0(Q) < \theta_0, \mu_1(Q) \geq \theta_1\}$ and $L^* - 2$ on $\{\mu_1(Q) < \theta_1\}$.

Step 4: decomposition. By Step 3, and treating an event that prescribes a negative depth as empty, the event $\{\text{level}_{\text{diff}}(Q) = k\}$ is the disjoint union of the following three events, listed according to the level returned by the c -2D-DAG (L^* , $L^* - 1$, $L^* - 2$). Each is the intersection of an event of dimension 0 (left) and an event of dimension 1 (right); we omit the argument Q .

Case 1, $k = 2j$: $\{\mu_0 \geq \kappa_0, \lambda_0 = \kappa_0 - j\} \cap \{\mu_1 \geq \kappa_0, \lambda_1 \geq \kappa_0 - j\}$; $\{\mu_0 \geq \kappa_0, \lambda_0 \geq \kappa_0 - j\} \cap \{\mu_1 < \kappa_0, \lambda_1 = \kappa_0 - j - 1\}$; $\{\mu_0 < \kappa_0, \lambda_0 = \kappa_0 - j - 1\} \cap \{\lambda_1 \geq \kappa_0 - j - 1\}$.

Case 1, $k = 2j + 1$: $\{\mu_0 \geq \kappa_0, \lambda_0 \geq \kappa_0 - j\} \cap \{\mu_1 \geq \kappa_0, \lambda_1 = \kappa_0 - j - 1\}$; $\{\mu_0 \geq \kappa_0, \lambda_0 = \kappa_0 - j - 1\} \cap \{\mu_1 < \kappa_0, \lambda_1 \geq \kappa_0 - j - 1\}$; $\{\mu_0 < \kappa_0, \lambda_0 \geq \kappa_0 - j - 1\} \cap \{\lambda_1 = \kappa_0 - j - 2\}$.

Case 2, $k = 2j$: $\{\mu_0 \geq \kappa_1 + 1, \lambda_0 \geq \kappa_1 - j + 1\} \cap \{\mu_1 \geq \kappa_1, \lambda_1 = \kappa_1 - j\}$; $\{\mu_0 < \kappa_1 + 1, \lambda_0 = \kappa_1 - j\} \cap \{\mu_1 \geq \kappa_1, \lambda_1 \geq \kappa_1 - j\}$; $\{\lambda_0 \geq \kappa_1 - j\} \cap \{\mu_1 < \kappa_1, \lambda_1 = \kappa_1 - j - 1\}$.

Case 2, $k = 2j + 1$: $\{\mu_0 \geq \kappa_1 + 1, \lambda_0 = \kappa_1 - j\} \cap \{\mu_1 \geq \kappa_1, \lambda_1 \geq \kappa_1 - j\}$; $\{\mu_0 < \kappa_1 + 1, \lambda_0 \geq \kappa_1 - j\} \cap \{\mu_1 \geq \kappa_1, \lambda_1 = \kappa_1 - j - 1\}$; $\{\lambda_0 = \kappa_1 - j - 1\} \cap \{\mu_1 < \kappa_1, \lambda_1 \geq \kappa_1 - j - 1\}$.

For instance, on the first event of Case 1 with $k = 2j$, the c -2D-DAG returns $2\kappa_0$ and, since $\lambda_0 = \kappa_0 - j \leq \lambda_1$, the 2D-Tree returns $2\kappa_0 - 2j$; conversely, if the c -2D-DAG returns $2\kappa_0$ and the 2D-Tree returns the even level $2\kappa_0 - 2j$, then $\lambda_0 = \kappa_0 - j \leq \lambda_1$. The other events are verified in the same way.

Step 5: probabilities. Since $(\lambda_0(Q), \mu_0(Q))$ and $(\lambda_1(Q), \mu_1(Q))$ are independent, each event in Step 4 has probability equal to the product of the probabilities of its two factors, which are given by Step 2 and (3). An equality $\lambda_i = \theta_i$ appears only for $k = 0$, and then $\theta_i = \kappa_i$; since $\lambda_i(Q) \leq \kappa_i$, the corresponding factor equals $\Pr[\lambda_i \geq \theta_i, \mu_i \geq \theta_i] = (N_i - 2^{\theta_i} s_i)/(N_i - s_i)$, which is why $k = 0$ is listed separately. For $k = L^*$, only the first event is nonempty, and for $k = L^* - 1$, the third event is empty; this gives the remaining separate rows. Multiplying the factors and clearing the common denominator $(N_0 - s_0)(N_1 - s_1)$ yields the stated formulas. $\square$

F Proof of Theorem 3.5 from Section 3.2

PROOF. By linearity of expectation, it suffices to bound $\mathbb{E}[\text{level}_{\text{diff}}(Q)]$ and $\mathbb{E}[AR(Q)]$ for a query Q drawn from $\mathcal{I}_s$; all probabilities and expectations are taken under $\mathcal{I}_s$. Let $m = \min\{\kappa_0, \kappa_1\} = \lfloor L^*/2 \rfloor$. If $L^* = 0$, then $\text{level}_{\text{diff}}(Q) = 0$ and $AR(Q) = 1$ with probability 1 (Lemma 3.4), and both claims hold, since then $m = 0$ and $N_0/s_0 < 2$. Assume $L^* \geq 1$, and let $r = \frac{c-2}{c-1}$, so that $\frac{(c-2)(5c-3)}{(c-1)^2} = 7r - 2r^2$. We use the following facts.

(F1) Tails. For X with values in $\{0, \dots, L^*\}$, $\mathbb{E}[X] = \sum_{\ell=1}^{L^*} \Pr[X \geq \ell]$. By Lemma 3.3(ii), $\Pr[\text{level}_{\text{c-DAG}}(Q) \geq \ell]$ equals 1 for $\ell \leq L^* - 2$, p_* for $\ell = L^* - 1$, and $p_0 p_1$ for $\ell = L^*$, where $p_* = p_0$ if L^* is even and $p_* = p_1$ otherwise. By Lemma 3.2(i) and (3), for $1 \leq \ell \leq L^*$, $1 - \Pr[\text{level}_{\text{Tree}}(Q) \geq \ell] = u + v - uv$ with $u = (2^{\lceil \ell/2 \rceil} - 1)s_0/(N_0 - s_0)$ and $v = (2^{\lfloor \ell/2 \rfloor} - 1)s_1/(N_1 - s_1)$. Hence

$$\mathbb{E}[\text{level}_{\text{diff}}(Q)] = \sum_{\ell=1}^{L^*} (u + v - uv) - (1 - p_*) - (1 - p_0 p_1). \quad (7)$$

(F2) Deficits. For $0 \leq d \leq \kappa_i$, $(2^d - 1)s_i/(N_i - s_i) \leq 2^d s_i/N_i$, since this is equivalent to $2^d s_i \leq N_i$. Moreover, $x + y - xy$ is nondecreasing in each of $x, y \in [0, 1]$.

(F3) Penalties. By Step 1 of the proof of Lemma 3.4, $1 - p_i = (2^{\theta_i} - 1)\eta_i/(N_i - s_i)$. Call dimension i critical if $1 \leq \theta_i = \kappa_i$; otherwise $\eta_i = 0$ and $p_i = 1$. In each case below, we fix $p_i \in (0, 1]$ such that $\rho_i = 2^{\kappa_i} s_i/N_i$ if dimension i is critical and $\rho_i \leq \frac{1}{2}$ otherwise, and set $q_i = (c-1)(\rho_i - r)^+$. If dimension i is critical, then $\eta_i = q_i N_i/2^{\kappa_i}$ and $N_i - s_i = (2^{\kappa_i} - \rho_i)N_i/2^{\kappa_i}$, hence $1 - p_i = (2^{\kappa_i} - 1)q_i/(2^{\kappa_i} - \rho_i)$; otherwise $q_i = 0 = 1 - p_i$, since $r \geq \frac{1}{2}$. In both cases,

$$(1 - 2^{-\kappa_i})q_i \leq 1 - p_i \leq q_i \leq 1.$$

(F4) Maximization. Let $H(x, y) = 4x + 3y - 2xy - 2q_x - (1 - q_x)q_y$ with $q_z = (c-1)(z-r)^+$. Then $H(x, y) \leq 7r - 2r^2$ for all $x \in [\frac{1}{2}, 1]$ and $y \in [0, 1]$. Indeed, for $x \geq r$, $\partial H/\partial x = 4 - 2y - (c-1)(2 - q_y)$; this is $4 - 2y - 2(c-1) \leq 0$ for $y \leq r$, and on $[r, 1]$ it is affine in y with values $4 - 2r - 2(c-1) \leq 0$ at $y = r$ and $3 - c \leq 0$ at $y = 1$. Hence $H(x, y) \leq H(\min\{x, r\}, y)$. For $x \leq r$, $H(x, y) = 4x + 3y - 2xy - q_y$, and $\partial H/\partial y = 3 - 2x > 0$ for $y < r$ and $\partial H/\partial y = 3 - 2x - (c-1) \leq 0$ for $y > r$, since $x \geq \frac{1}{2}$ and $c \geq 3$. Hence $H(x, y) \leq H(x, r) = 4x + 3r - 2xr \leq H(r, r)$.

(i), Case 1: $\kappa_0 \leq \kappa_1$, so that $L^* = 2\kappa_0$ and $p_* = p_0$. Let $K = 2^{\kappa_0}$, $\rho_0 = Ks_0/N_0 \in (\frac{1}{2}, 1]$, and $\rho_1 = Ks_1/N_1 \in (0, 1]$.

Dimension 0 is critical, and dimension 1 is critical if and only if $\kappa_1 = \kappa_0$; otherwise $\rho_1 \leq 2^{\kappa_0 - \kappa_1} \leq \frac{1}{2}$, so (F3) applies, with $2^{\kappa_1} = K$ whenever dimension 1 is critical. For $j \in \{0, \dots, \kappa_0 - 1\}$, by (F2), the levels $\ell = 2(\kappa_0 - j)$ and $\ell = 2(\kappa_0 - j) - 1$ have $u \leq 2^{-j}\rho_0$, and $v \leq 2^{-j}\rho_1$ and $v \leq 2^{-j-1}\rho_1$, respectively. Hence

$$\begin{aligned} \sum_{\ell=1}^{L^*} (u + v - uv) &\leq \sum_{j=0}^{\kappa_0-1} \left[(2\rho_0 + \tfrac{3}{2}\rho_1)2^{-j} - \tfrac{3}{2}\rho_0\rho_1 4^{-j} \right] \\ &= (1 - \tfrac{1}{K})(4\rho_0 + 3\rho_1) - 2(1 - \tfrac{1}{K^2})\rho_0\rho_1 \\ &\leq (1 - \tfrac{1}{K})(4\rho_0 + 3\rho_1 - 2\rho_0\rho_1). \end{aligned}$$

By (F3) and $p_0 \geq 1 - q_0$, $(1 - p_0) + (1 - p_0 p_1) = 2(1 - p_0) + p_0(1 - p_1) \geq (1 - \frac{1}{K})(2q_0 + (1 - q_0)q_1)$. By (7) and (F4), $\mathbb{E}[\text{level}_{\text{diff}}(Q)] \leq (1 - \frac{1}{K})H(\rho_0, \rho_1) < 7r - 2r^2$.

(i), Case 2: $\kappa_0 > \kappa_1$, so that $L^* = 2\kappa_1 + 1$ and $p_* = p_1$. If $\kappa_1 = 0$, then $\mathbb{E}[\text{level}_{\text{diff}}(Q)] \leq L^* = 1 < 3 \leq 7r - 2r^2$. Otherwise, let $K = 2^{\kappa_1}$, $\rho_0 = 2Ks_0/N_0 \in (0, 1]$, and $\rho_1 = Ks_1/N_1 \in (\frac{1}{2}, 1]$. Dimension 1 is critical, and dimension 0 is critical if and only if $\kappa_0 = \kappa_1 + 1$; otherwise $\rho_0 \leq 2^{\kappa_1+1-\kappa_0} \leq \frac{1}{2}$, so (F3) applies, with $2^{\kappa_0} = 2K$ whenever dimension 0 is critical. For $j \in \{0, \dots, \kappa_1\}$, the level $\ell = 2(\kappa_1 - j) + 1$ has $u \leq 2^{-j}\rho_0$ and $v \leq 2^{-j}\rho_1$; for $j \in \{0, \dots, \kappa_1 - 1\}$, the level $\ell = 2(\kappa_1 - j)$ has $u \leq 2^{-j-1}\rho_0$ and $v \leq 2^{-j}\rho_1$. Summing as in Case 1,

$$\sum_{\ell=1}^{L^*} (u + v - uv) \leq 3\rho_0 + 4\rho_1 - 2\rho_0\rho_1 - \frac{2\rho_0 + 3\rho_1}{K} + \frac{\rho_0\rho_1}{K^2}.$$

By (F3) and $p_1 \geq 1 - q_1$, $(1 - p_1) + (1 - p_0 p_1) = 2(1 - p_1) + p_1(1 - p_0) \geq 2q_1 + (1 - q_1)q_0 - \frac{1}{K}(2q_1 + \frac{1}{2}q_0)$. By (7),

$$\mathbb{E}[\text{level}_{\text{diff}}(Q)] \leq H(\rho_1, \rho_0) - \frac{B}{K},$$

where $B \geq 2\rho_0 + 3\rho_1 - \frac{1}{2}\rho_0\rho_1 - 2q_1 - \frac{1}{2}q_0$. Since $q_z \leq 2z - 1 \leq z$ whenever $q_z > 0$, we get $B \geq \frac{3}{2}\rho_0 + \rho_1 - \frac{1}{2}\rho_0\rho_1 > 0$, and (F4) gives $\mathbb{E}[\text{level}_{\text{diff}}(Q)] < 7r - 2r^2$.

(ii). By Section 3, $AR(Q) = 2^{\text{level}_{\text{diff}}(Q)} \geq 1$. We may assume $m \geq 1$, since otherwise $2^{m-1} - (m+2)/4 \leq 0$.

Case 1: $\kappa_0 \leq \kappa_1$, so that $m = \kappa_0$. Fix $a \in \{0, \dots, m-1\}$. On the event $\{\lambda_0(Q) = a \leq \lambda_1(Q)\}$, the 2D-Tree returns level $2a$ (Lemma 3.2(i)). By Lemma 3.3(i) and $\mu_1(Q) \geq \kappa_1 - 1 \geq m-1$, the c -2D-DAG returns at least $2m-1$ if $\mu_0(Q) \geq m$, and $2m-2$ otherwise. By Step 2 of the proof of Lemma 3.4, with $\theta_0 = \kappa_0$, $\Pr[\lambda_0(Q) = a, \mu_0(Q) \geq m] = 2^a(s_0 - \eta_0)/(N_0 - s_0)$ and $\Pr[\lambda_0(Q) = a, \mu_0(Q) < m] = 2^a\eta_0/(N_0 - s_0)$. By independence,

$$\mathbb{E}[AR(Q)] \geq \sum_{a=0}^{m-1} 2^{2m-2-2a} \frac{2^a(2s_0 - \eta_0)}{N_0 - s_0} \Pr[\lambda_1(Q) \geq a].$$

We have $2s_0 - \eta_0 \geq N_0/2^m$: if $\eta_0 = 0$, since $s_0 > N_0/2^{m+1}$; otherwise, $2s_0 - \eta_0 = (c-2)N_0/2^m - (c-3)s_0 \geq N_0/2^m$, since $s_0 \leq N_0/2^m$. Hence $(2s_0 - \eta_0)/(N_0 - s_0) > 2^{-m}$. By (3) and (F2), $\Pr[\lambda_1(Q) \geq a] \geq 1 - 2^a s_1/N_1 \geq 1 - 2^{a-m}$, since $s_1 \leq N_1/2^{\kappa_1} \leq N_1/2^m$. Therefore

$$\mathbb{E}[AR(Q)] > \sum_{a=0}^{m-1} 2^{2m-2-a} (1 - 2^{a-m}) = 2^{m-1} - \frac{m+2}{4}.$$

Case 2: $\kappa_0 > \kappa_1$, so that $m = \kappa_1$. The argument is the same with the roles of the dimensions exchanged: for $a \in \{0, \dots, m-1\}$, on $\{\lambda_1(Q) = a < \lambda_0(Q)\}$ the 2D-Tree returns level $2a+1$, the c -2D-DAG returns at least $2m$ if $\mu_1(Q) \geq m$ (since $\mu_0(Q) \geq \kappa_0 - 1 \geq m$) and $2m-1$ otherwise, $2s_1 - \eta_1 \geq N_1/2^m$, and $\Pr[\lambda_0(Q) \geq a+1] \geq 1 - 2^{a+1}s_0/N_0 \geq 1 - 2^{a-m}$, since $\kappa_0 \geq m+1$. This yields the same sum.

Finally, $2^{\kappa_i} > N_i/(2s_i)$ by the definition of κ_i , so $2^{m-1} > \frac{1}{4} \min\{N_0/s_0, N_1/s_1\}$.

Tightness of (i). If $\kappa_0 = \kappa_1 = \kappa$ and $s_i = rN_i/2^\kappa$, then $\eta_0 = \eta_1 = 0$, so $p_0 = p_1 = 1$, and the bounds of (F2) are exact up to terms that vanish as $\kappa \rightarrow \infty$; hence $\mathbb{E}[\text{level}_{\text{diff}}(Q)] \rightarrow H(r, r) = 7r - 2r^2$. $\square$

Exact values. The proof also yields the exact expectations, which are used in Figure 4. By (F1), with p_0, p_1 as in Lemma 3.3(ii), $\mathbb{E}[\text{level}_{\text{diff}}(Q)] = L^* - 2 + p_* + p_0 p_1 - \mathbb{E}[\text{level}_{\text{Tree}}(Q)]$, where $(N_0 - s_0)(N_1 - s_1) \mathbb{E}[\text{level}_{\text{Tree}}(Q)]$ equals

$$\begin{aligned} &2\kappa_0 N_0 N_1 - 3(2^{\kappa_0} - 1)N_0 s_1 - 4(2^{\kappa_0} - 1)N_1 s_0 \\ &+ 2(4^{\kappa_0} - 1)s_0 s_1 \quad \text{in Case 1, and} \\ &(2\kappa_1 + 1)N_0 N_1 - (2^{\kappa_1+2} - 3)N_0 s_1 - (3 \cdot 2^{\kappa_1+1} - 4)N_1 s_0 \\ &+ (4^{\kappa_1+1} - 2)s_0 s_1 \quad \text{in Case 2.} \end{aligned}$$

Summing 2^k times the probabilities of Lemma 3.4, which are geometric in j , gives $(N_0 - s_0)(N_1 - s_1) \mathbb{E}[AR(Q)]$; with $K = 2^{\kappa_0}$ in Case 1, it equals

$$\begin{aligned} &N_0 N_1 - 3KN_1 s_0 - 2KN_0 s_1 \\ &+ K^2(2N_1 s_0 + N_0 s_1 + (1 - 2\kappa_0)s_0 s_1) \\ &- K\eta_0 \left[\tfrac{3}{4}(K-1)(2N_1 + Ks_1) - \tfrac{3}{2}\kappa_0 Ks_1 \right] \\ &- K\eta_1 \left[\tfrac{1}{2}(K-1)N_0 + (K - \kappa_0 - 1)Ks_0 \right] \\ &+ K^2\eta_0\eta_1 \left[\tfrac{3}{2}(K-1) - \kappa_0 \right], \end{aligned}$$

and, with $K = 2^{\kappa_1}$ in Case 2,

$$\begin{aligned} &N_0 N_1 - 4KN_1 s_0 - 3KN_0 s_1 \\ &+ K^2(4N_1 s_0 + 2N_0 s_1 - 4\kappa_1 s_0 s_1) \\ &- K\eta_0 \left[(2K-1)N_1 + (2K - 2\kappa_1 - 3)Ks_1 \right] \\ &- 3K\eta_1 \left[\tfrac{1}{2}(K-1)N_0 + (K - \kappa_1 - 1)Ks_0 \right] \\ &+ 2K^2\eta_0\eta_1(2K - \kappa_1 - 2). \end{aligned}$$

G Proof of Corollary 3.6 from Section 3.2

PROOF. Fix $t \in \{1, \dots, \tau\}$, and let J_t denote the index of the shape selected by Q_t , so that $\Pr[J_t = j] = \omega_j$ and, given $J_t = j$, the query Q_t is drawn from $\mathcal{I}_{s_j}$. By the law of total expectation and Theorem 3.5 (with $\tau = 1$ and shape s_j),

$$\mathbb{E}[\text{level}_{\text{diff}}(Q_t)] = \sum_{j=1}^{\sigma} \omega_j \mathbb{E}_{\mathcal{I}_{s_j}}[\text{level}_{\text{diff}}(Q)] < \frac{(c-2)(5c-3)}{(c-1)^2},$$

since the bound of Theorem 3.5(i) does not depend on the shape. If instead Q_t is drawn from $\mathcal{I}_{s(t)}$ for a fixed shape $s(t)$ satisfying the shape assumptions, the same bound holds for $\mathbb{E}_{\mathcal{I}_{s(t)}}[\text{level}_{\text{diff}}(Q)]$ directly. In both cases, averaging over t and using linearity of expectation proves (i).

Similarly, by Theorem 3.5(ii) and $\min\{\kappa_j^{(0)}, \kappa_j^{(1)}\} = m_j$,

$$\mathbb{E}[AR(Q_t)] = \sum_{j=1}^{\sigma} \omega_j \mathbb{E}_{\mathcal{I}_{s_j}}[AR(Q)] \geq \sum_{j=1}^{\sigma} \omega_j \max\{1, g(m_j)\},$$

where $g(x) = 2^{x-1} - (x+2)/4$; averaging over t gives the first inequality of (ii). Since $\max\{1, g(m_j)\}$ is at least 1 and at least $g(m_j)$, and the ω_j sum to 1, $\sum_j \omega_j \max\{1, g(m_j)\} \geq \max\{1, \sum_j \omega_j g(m_j)\}$. Finally, g is convex on $\mathbb{R}$, as the sum of a convex exponential and an affine function, so Jensen's inequality gives $\sum_j \omega_j g(m_j) \geq g(\bar{m})$, which proves the second inequality.

For the bound stated after the corollary, with Γ as defined there: by the definition of $\kappa_j^{(i)}$, $m_j > \log_2 \min\{N_0/s_j^{(0)}, N_1/s_j^{(1)}\} - 1$, hence $\bar{m} > \log_2 \Gamma - 1$. Since $g'(x) = 2^{x-1} \ln 2 - \frac{1}{4} \geq \frac{1}{2} \ln 2 - \frac{1}{4} > 0$

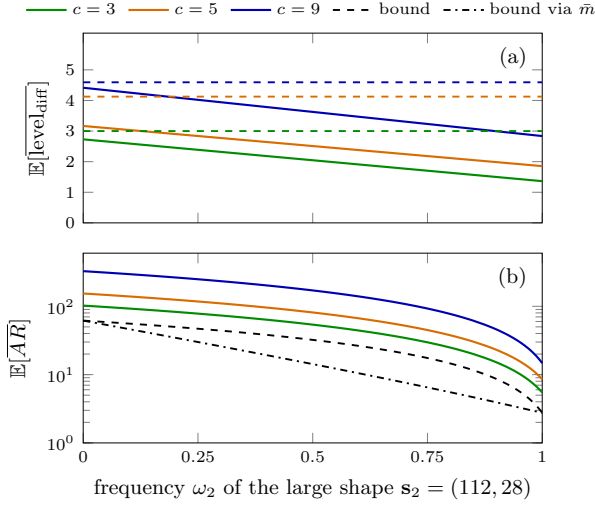


Figure 11: Illustration of Corollary 3.6 for $N_0 = N_1 = 1024$, $c \in \{3, 5, 9\}$, and a stream mixing the small square shape $s_1 = (7, 7)$ ($m_1 = 7$) and the large rectangular shape $s_2 = (112, 28)$ ($m_2 = 3$) with frequencies $\omega_1 = 1 - \omega_2$ and ω_2 . Solid lines are exact values. (a) $\mathbb{E}[\text{level}_{\text{diff}}]$ and, dashed in the same colour, the bound of part (i). (b) $\mathbb{E}[\overline{AR}]$ (log scale), the first bound of part (ii) (dashed), and the second bound of part (ii), via $\bar{m} = 7 - 4\omega_2$ (dash-dotted).

for $x \geq 0$, g is increasing on $[0, \infty)$; moreover, $\bar{m} \geq 0$ and $\Gamma > 1$, since $s_j^{(i)} < N_j$. Hence either $\log_2 \Gamma \geq 1$ and $g(\bar{m}) > g(\log_2 \Gamma - 1) = (\Gamma - \log_2 \Gamma - 1)/4$, or $1 < \Gamma < 2$ and $(\Gamma - \log_2 \Gamma - 1)/4 \leq 0$, since $\log_2 \Gamma \geq \Gamma - 1$ on $[1, 2]$ by concavity. In both cases, $\mathbb{E}[\overline{AR}] \geq \max\{1, (\Gamma - \log_2 \Gamma - 1)/4\}$. $\square$

Corollary 3.6 extends the asymmetry of Theorem 3.5 to mixed streams (Figure 11). Observe that both exact measures are linear in ω_2 , being mixtures of the two single-shape values, and decrease as the large shape becomes more frequent: for $c = 3$, $\mathbb{E}[\text{level}_{\text{diff}}]$ drops from 2.73 to 1.36 and $\mathbb{E}[\overline{AR}]$ from 102.4 to 5.5. The level-overhead bound is shape-agnostic, so mixing query shapes costs nothing in part (i). The accuracy guarantee is a frequency-weighted average of the single-shape guarantees: its first form is linear in ω_2 as well, whereas the second one depends on the shapes only through $\bar{m}$ and is convex, and therefore lies below it (Jensen's inequality). Let $\Gamma = \prod_{j=1}^{\sigma} \min\{N_0/s_j^{(0)}, N_1/s_j^{(1)}\}^{\omega_j}$ be the ω -weighted geometric mean of $\min\{N_0/s_j^{(0)}, N_1/s_j^{(1)}\}$. Since $m_j > \log_2 \min\{N_0/s_j^{(0)}, N_1/s_j^{(1)}\} - 1$, we have $\bar{m} > \log_2 \Gamma - 1$, and since $g(x) = 2^{x-1} - (x+2)/4$ is increasing on $[0, \infty)$, for $\Gamma \geq 2$

$$\mathbb{E}[\overline{AR}] \geq g(\bar{m}) > g(\log_2 \Gamma - 1) = \frac{\Gamma - \log_2 \Gamma - 1}{4},$$

while for $1 < \Gamma < 2$ the right-hand side is at most 1 and the bound $\max\{1, \cdot\}$ is trivial. In one dimension, the corresponding guarantee depends on the average of $\log(N/s_j)$; in two dimensions, it grows linearly in Γ , i.e., exponentially in the average of $\log_2 \min\{N_0/s_j^{(0)}, N_1/s_j^{(1)}\}$.

H Experimental Evaluation – More Details

This appendix completes Section 5: it reports the full table for the four query shapes, the counterparts of Figs. 5, 6, and 8 for the remaining shapes $(10^\circ, 10^\circ)$ and $(25^\circ, 20^\circ)$, and the root-return analysis used in Section 5.2.

Structures and SRC-search. We build the 2D-Tree and the 3-2D-DAG on D exactly as in Section 2.1: splits at the median of the distinct coordinates, alternating dimensions starting with latitude. Both are implemented in Python. SRC-search is the exhaustive search of [15], which returns a node minimizing $|D_o|$; ties go to the deeper node.

H.1 Remaining Shapes

Returned levels. The mean returned level is 7.77 against 5.87 for $(10^\circ, 10^\circ)$ and 5.29 against 4.43 for $(25^\circ, 20^\circ)$ (Fig. 12). As in Section 5.2, the CDF of the 3-2D-DAG never lies above that of the 2D-Tree, and smaller queries are answered at deeper levels.

Level Overhead Distribution. For $(10^\circ, 10^\circ)$ the baseline reproduces the shape of the LOD ($L_2 = 0.099$, nominal parameters 0.102). For $(25^\circ, 20^\circ)$ the fit is poor ($L_2 = 0.316$): the empirical LOD has mass 0.73 at $k = 0$ against 0.49 for the baseline, and a tail up to $k = 13$, whereas the baseline ends at $L^* = 4$ (Fig. 13). Part of the mass at $k = 0$ is the root-return mechanism of Appendix H.2: for $(25^\circ, 20^\circ)$ both structures return the root on the same 22.34% of queries.

Stream measures. The average Returned Level Overhead is 1.90 and 0.87, within 0.39 of the baseline expectation (1.86 and 1.26). The average Accuracy Ratio is 231.7 and 21.2; it exceeds both the baseline expectation (11.7 and 4.1) and the lower bound of Theorem 3.5(ii). The median is 2 and 1, and the geometric mean 3.72 and 1.83. Queries with $k > L^*$ are 3.8% and 6.4% of the stream and carry 95% and 91% of the sum of AR values; without them the mean drops to 12.5 and 2.0, of the order of the baseline expectations. Across the three smaller shapes, the queries with $AR \geq 1000$ contain on average only 18 to 34 elements, against 2,306 to 5,296 over all queries.

Stabilization. The L_2 distance between consecutive LODs falls below γ after 9,500 and 13,000 non-empty queries for the two shapes, and continues to decrease to about 10^{-4} at the end of the streams (Fig. 14).

Table 3: Gowalla dataset, $N = 2^{20}$: non-empty query streams of the four shapes (latitude $\times$ longitude, in degrees), referred to in Section 5.2. Columns: number of queries; query count at stabilization; mean returned levels; $\text{level}_{\text{diff}}$, empirical and baseline expectation (the bound of Theorem 3.5(i) is 3); mean, median, and geometric mean of AR ; expectation of AR under the fitted baseline and the lower bound of Theorem 3.5(ii); L_2 distance of the best-fit baseline LOD.

Shape	Queries	Stable at	mean level		$\text{level}_{\text{diff}}$		AR			$\mathbb{E}[AR]$		L_2
			Tree	DAG	emp.	base.	mean	med.	geo.	base.	l. bd.	
(5, 5)	138,500	18,500	6.75	9.45	2.69	2.17	305.9	4	6.46	25.4	14.25	0.057
(5, 10)	136,000	16,000	6.50	9.10	2.60	2.39	355.9	4	6.07	23.8	14.25	0.085
(10, 10)	129,500	9,500	5.87	7.77	1.90	1.86	231.7	2	3.72	11.7	6.5	0.099
(25, 20)	133,000	13,000	4.43	5.29	0.87	1.26	21.2	1	1.83	4.1	1.0	0.316

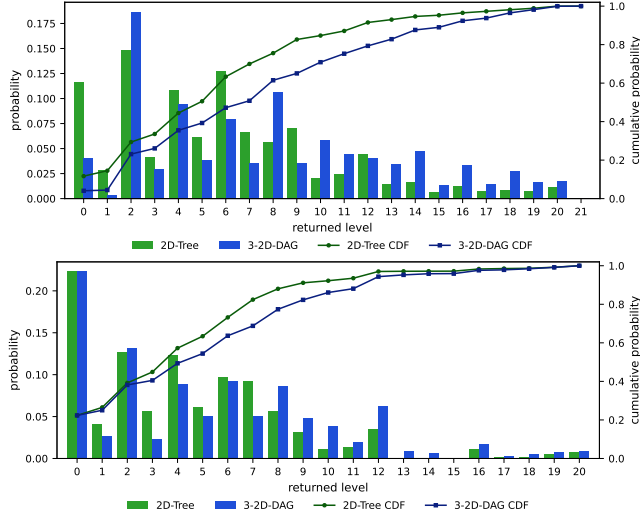


Figure 12: Returned level distributions on Gowalla, for the query shapes $(10^\circ, 10^\circ)$ and $(25^\circ, 20^\circ)$ (latitude $\times$ longitude); counterpart of Fig. 5. The 2D-Tree returns the root for 11.7% and 22.3% of the queries (Appendix H.2).

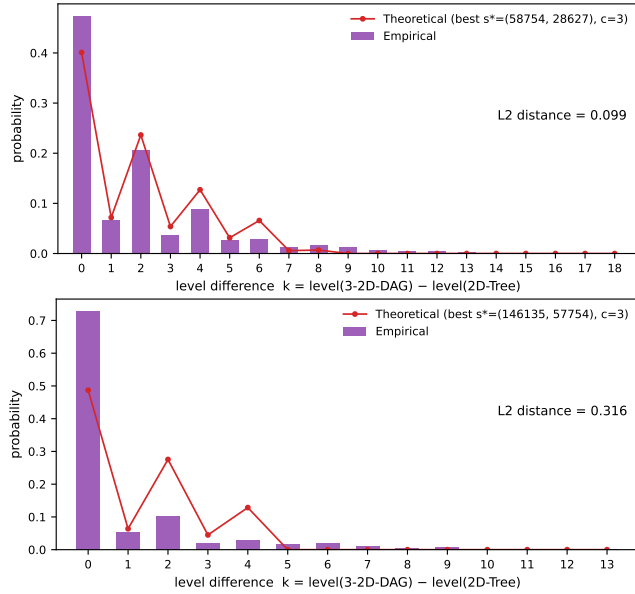


Figure 13: Level Overhead Distributions on Gowalla, for the query shapes $(10^\circ, 10^\circ)$ and $(25^\circ, 20^\circ)$; counterpart of Fig. 6.

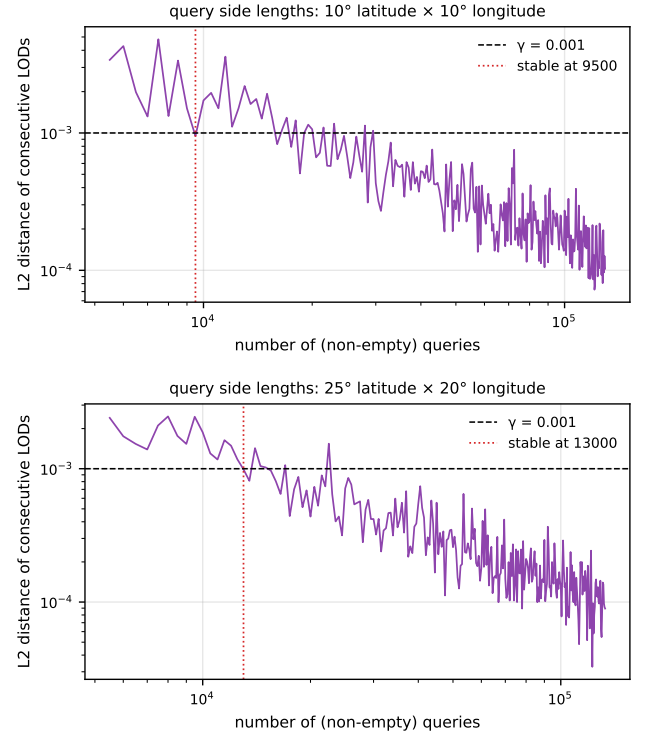


Figure 14: L_2 distance between the LODs of consecutive snapshots against the number of non-empty queries, for the query shapes $(10^\circ, 10^\circ)$ and $(25^\circ, 20^\circ)$; counterpart of Fig. 8.

H.2 Root Returns

The 2D-Tree returns its root for every query that is contained in neither child of the root. The root splits the latitudes of D at their median 40.344° , so the 2D-Tree returns the root exactly when this latitude lies in the interior of $Q^{(0)}$. For a uniform query this has probability $s_0/(180^\circ - s_0)$ (2.9% for $s_0 = 5^\circ$), but the median lies in a densely populated band, so among non-empty queries it is much larger: 7.60%, 6.68%, 11.67%, and 22.34% for the four shapes (leftmost bars of the 2D-Tree in Fig. 5). This condition predicts the returned root correctly for every one of the 537,000 queries.

The 3-2D-DAG has an additional child along longitude that spans the ranks between the first and the third quartile of the latitudes, i.e., $[33.83^\circ, 50.67^\circ)$ of width 16.85° . A query that straddles the median is contained in it if and only if it lies in this interval. Hence the 3-2D-DAG descends below the root for every straddling 5°

query and for some of the straddling 10° queries, but for none of the 25° queries, since they are wider than this child. It returns the root for 0%, 0%, 4.07%, and 22.34% of the queries, again in agreement with the condition on every query. For $(25^\circ, 20^\circ)$ both structures return the root on the same queries, which gives the 22.3% mass at level 0 in Fig. 5 and 22.3 of the 73 percentage points at $k = 0$ in Fig. 6.

This is the effect behind the asymmetry of Section 3.2: a root return costs the 2D-Tree a factor of order 2^{L^*} in the uniform model. On Gowalla, however, the queries that cause it lie in the dense band, where the 3-2D-DAG itself returns a shallow node (level 3.0 on average for the two 5° shapes). Their mean AR is only 15.7, 13.1, 3.4, and 1.0, and they contribute at most 1.1% of the sum of all AR values. The large values of AR below arise elsewhere (Section 5.2).

I Proof of Proposition 6.1 from Section 6

PROOF OF PROPOSITION 6.1. (i) follows from Section 2.1: the nodes containing Q form a path along which $|D_v|$ strictly decreases, so the minimizer is unique. (ii) Under these assumptions all nodes of $M(Q)$ lie on one level (Section 2.1), and by Lemma 3.1 the nodes of that level are products of one canonical interval per dimension, with the same depth in each dimension for all nodes of the level. In each dimension, the depth- d canonical intervals of Section 3 have a common length L and left endpoints spaced $L/(c-1)$ apart. An interval $[x, x+s)$ with $s > 0$ is contained in those whose left endpoint lies in a window of length $L-s < (c-1) \cdot L/(c-1)$, hence in at most $c-1$ of them. Therefore at most $(c-1)^2$ nodes of the level contain Q . (iii) is the example of Fig. 2. $\square$